

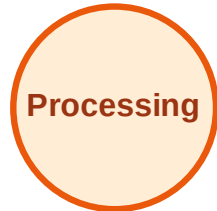
BFS Traversal

Step 1: Start at A

Legend



Complete



Processing



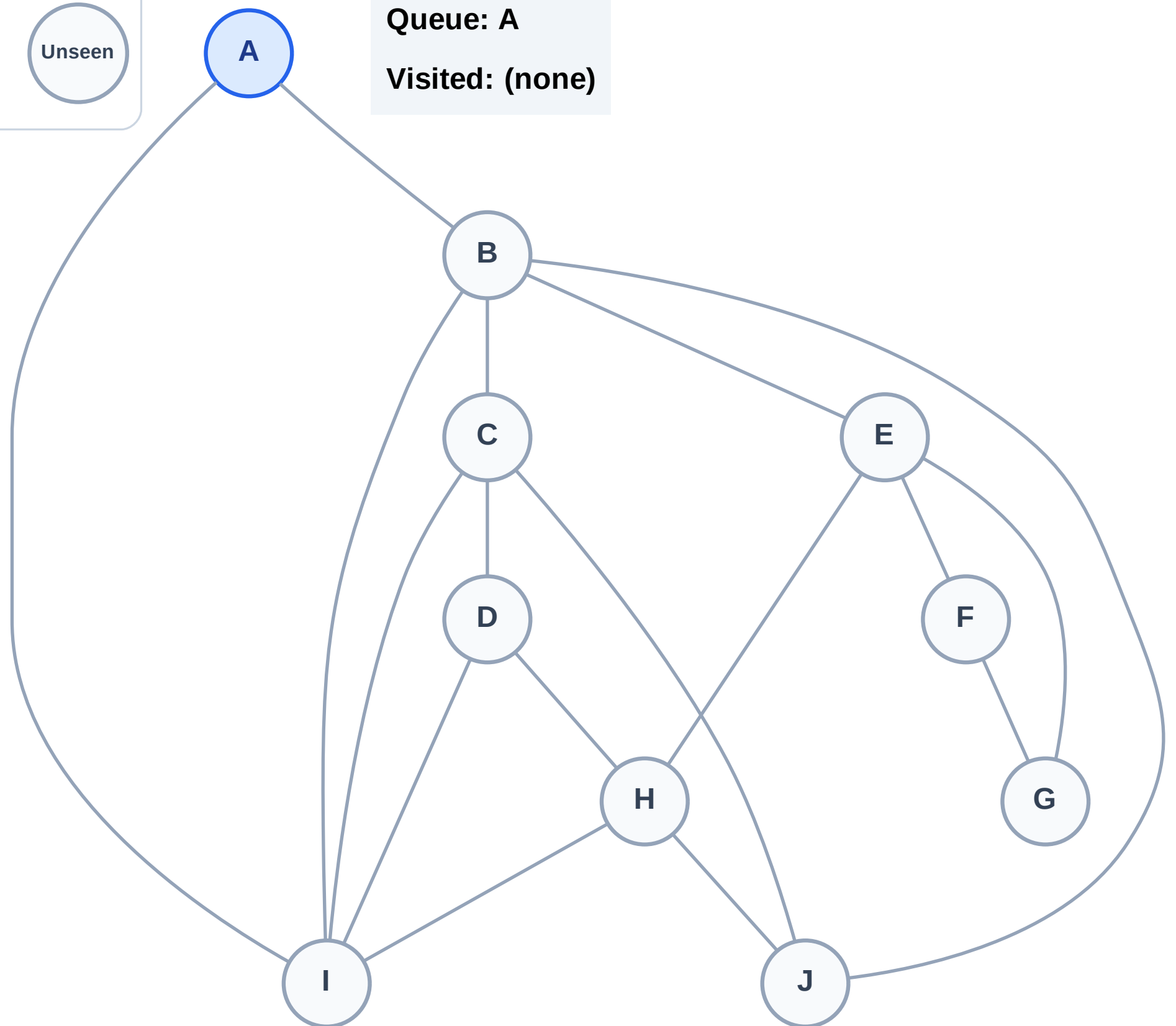
Discovered



Unseen

Queue: A

Visited: (none)



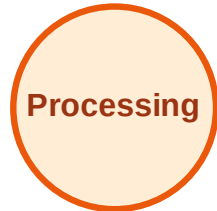
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



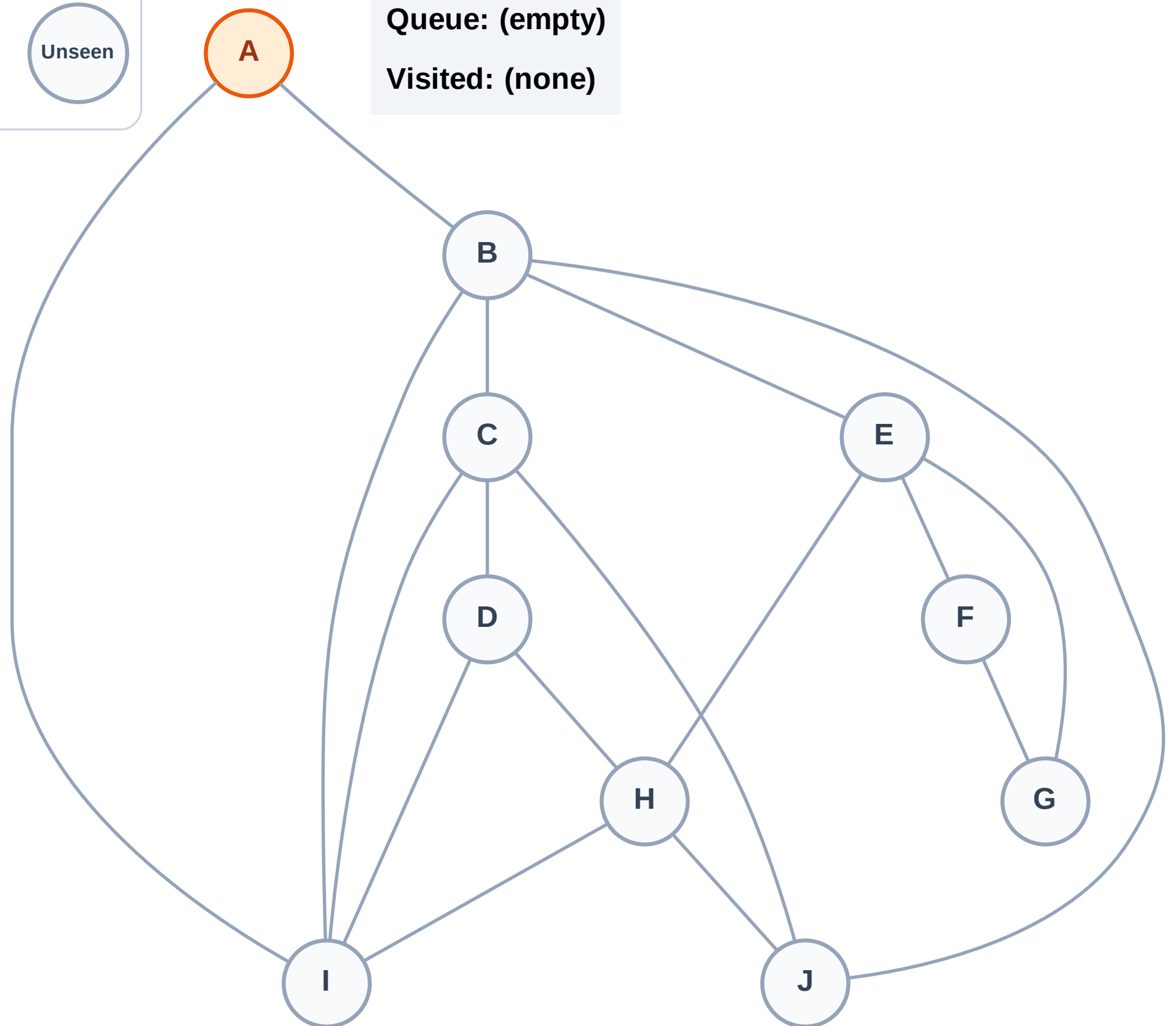
Discovered



Unseen

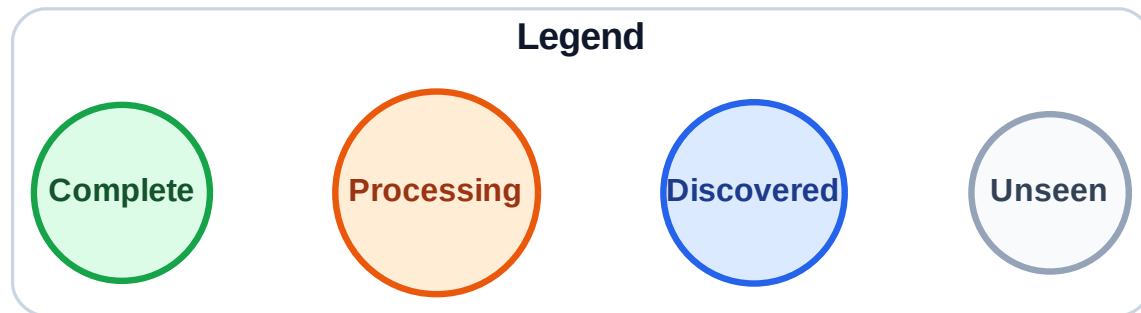
Queue: (empty)

Visited: (none)

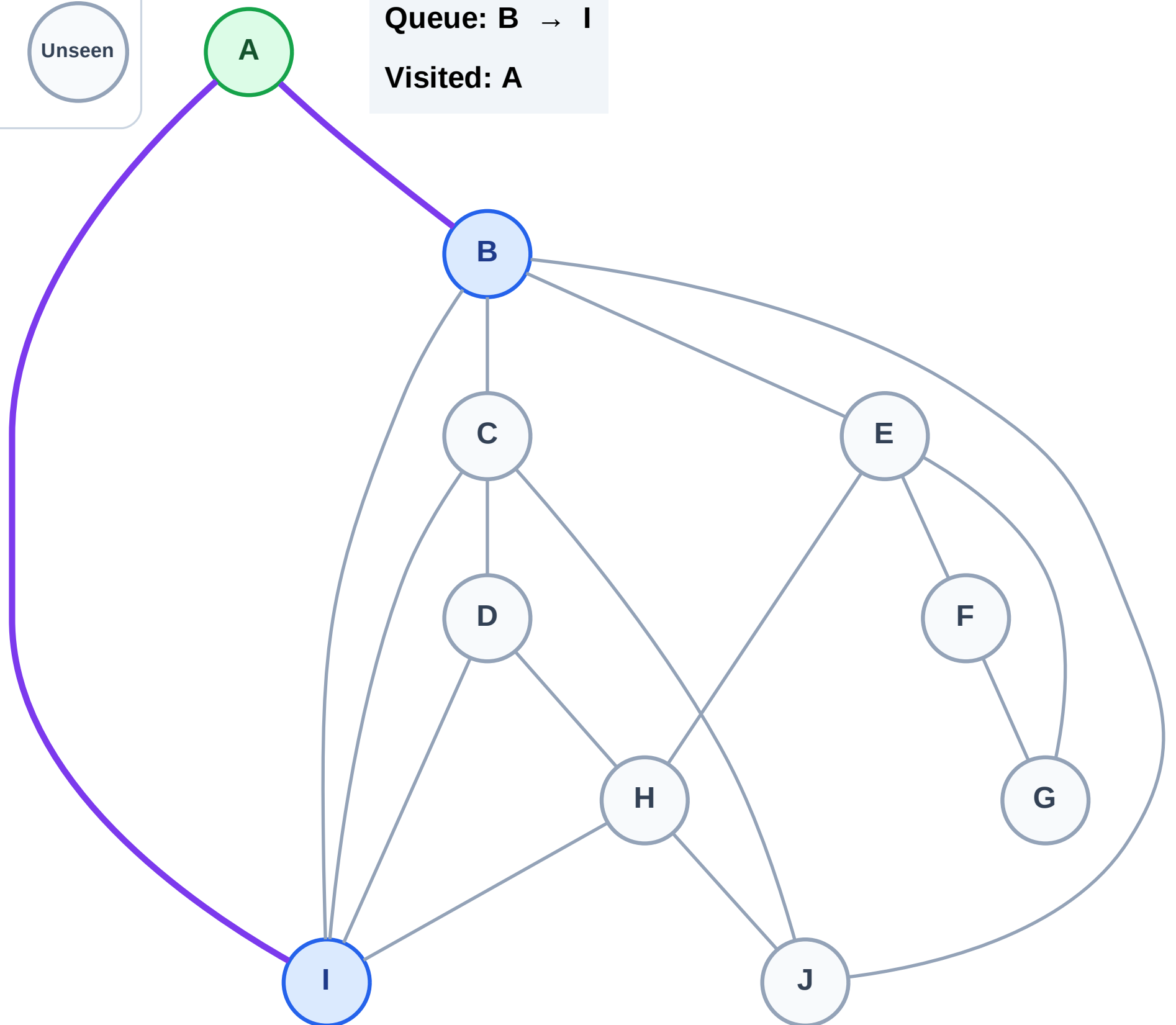


BFS Traversal

Step 3: Discover B, I from A



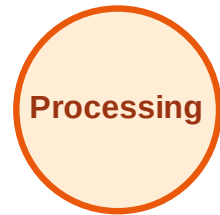
Queue: B → I
Visited: A



BFS Traversal

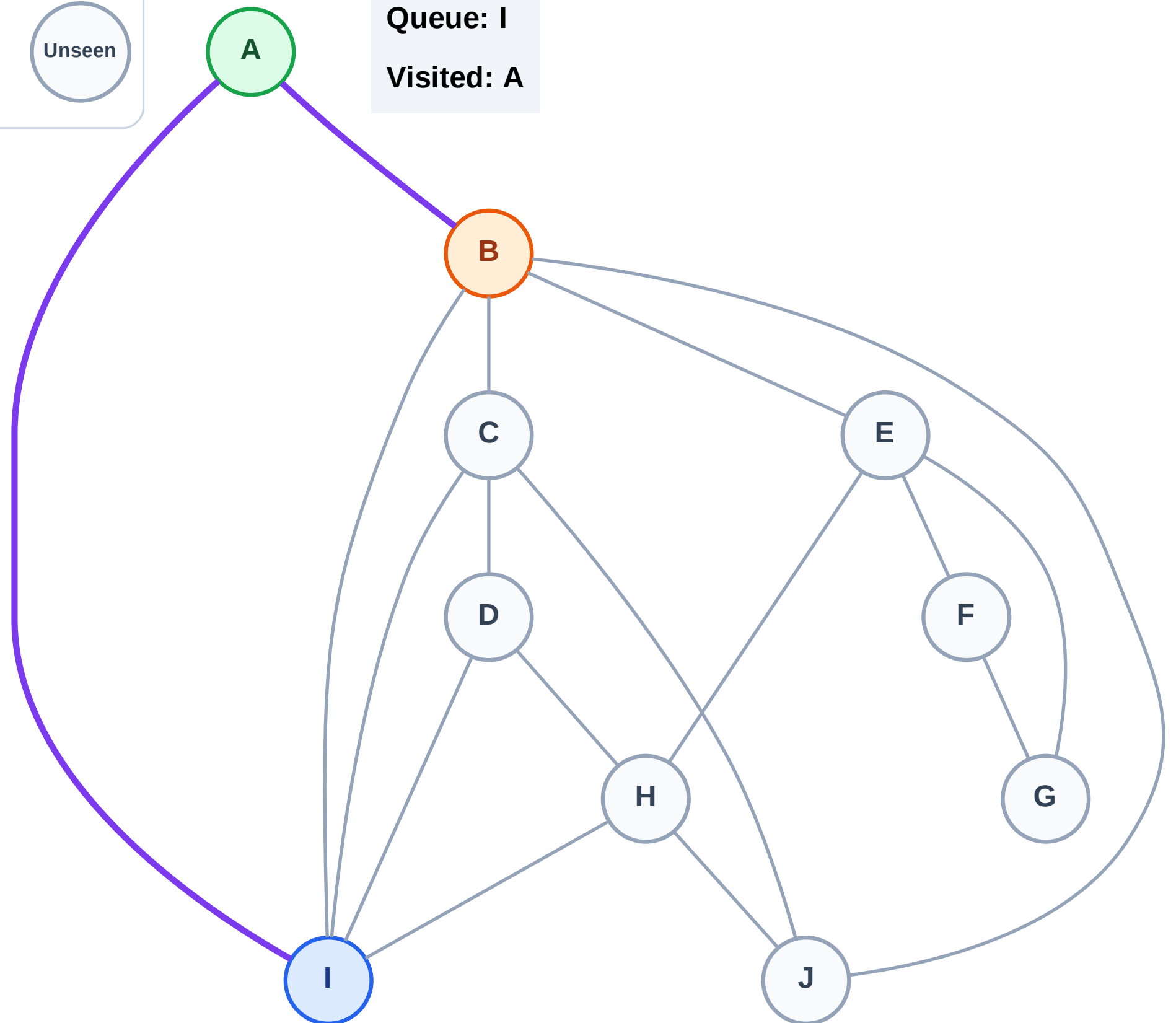
Step 4: Process node B

Legend



Queue: I

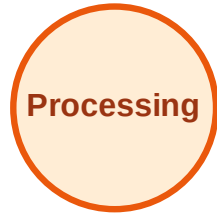
Visited: A



BFS Traversal

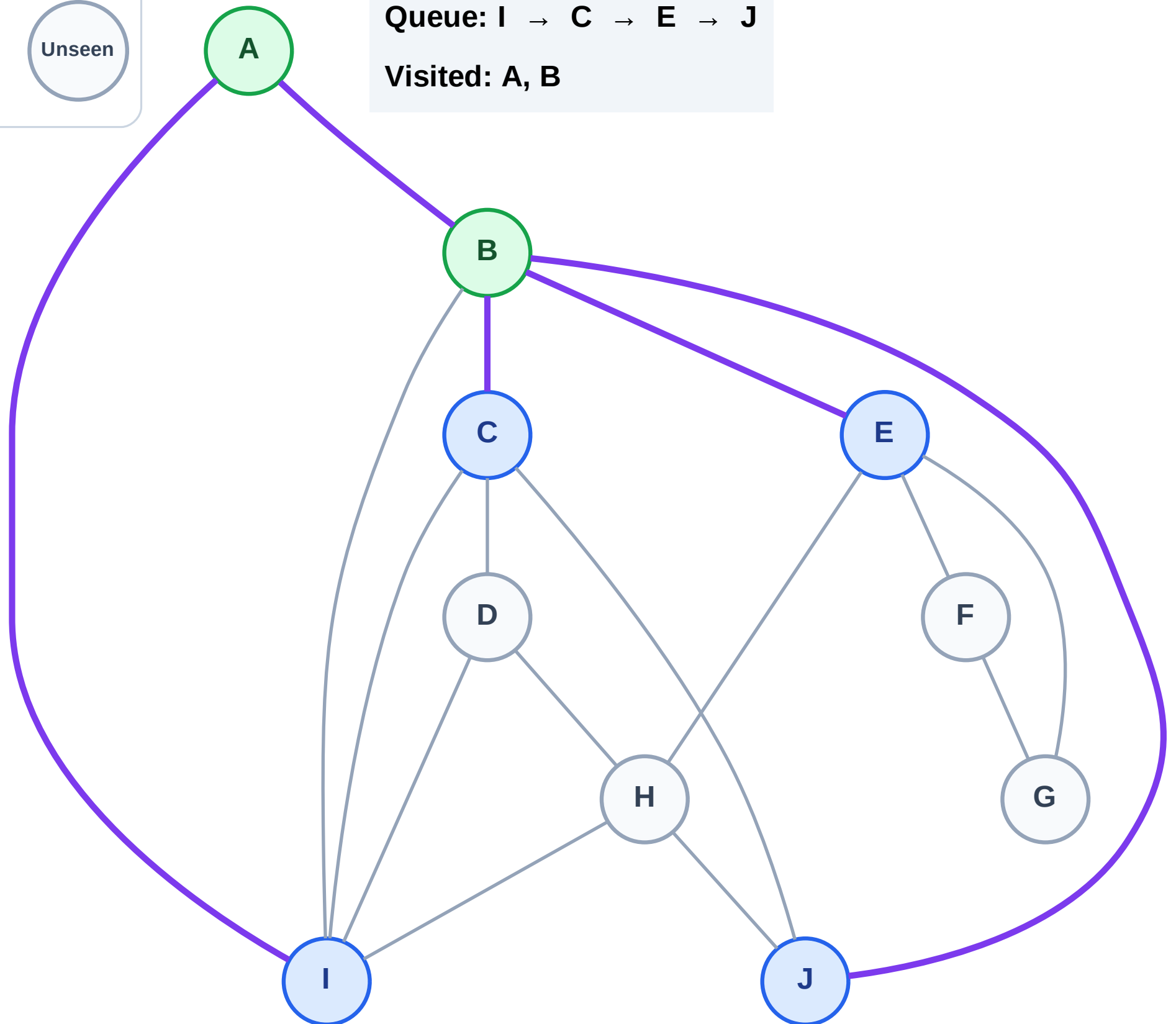
Step 5: Discover C, E, J from B

Legend



Queue: I → C → E → J

Visited: A, B



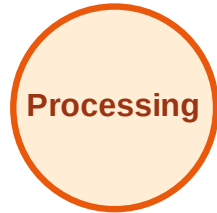
BFS Traversal

Step 6: Process node I

Legend



Complete



Processing



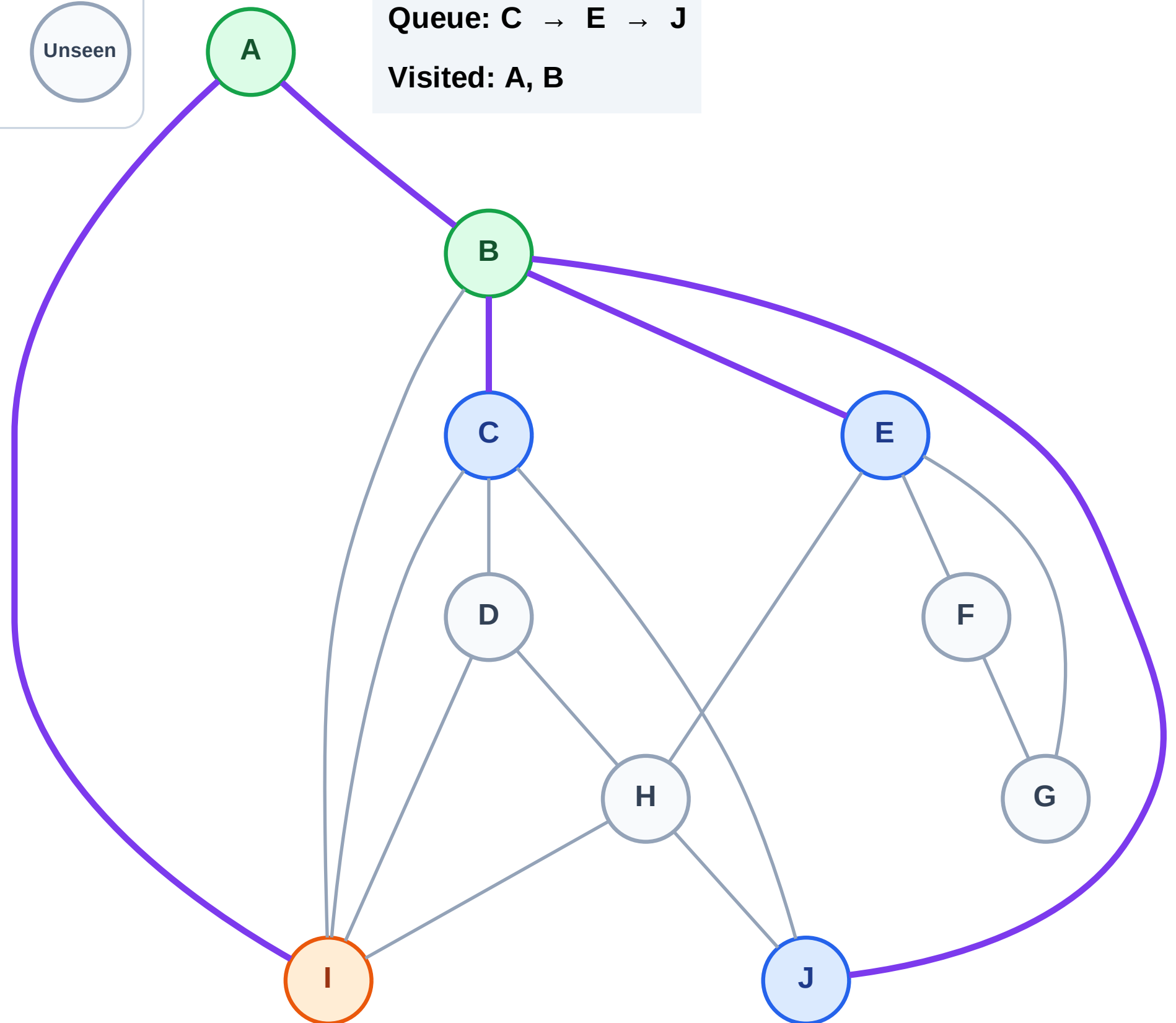
Discovered



Unseen

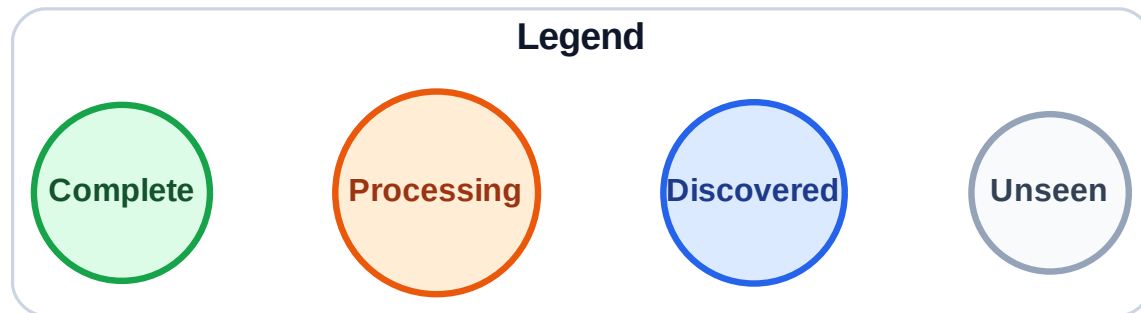
Queue: C → E → J

Visited: A, B



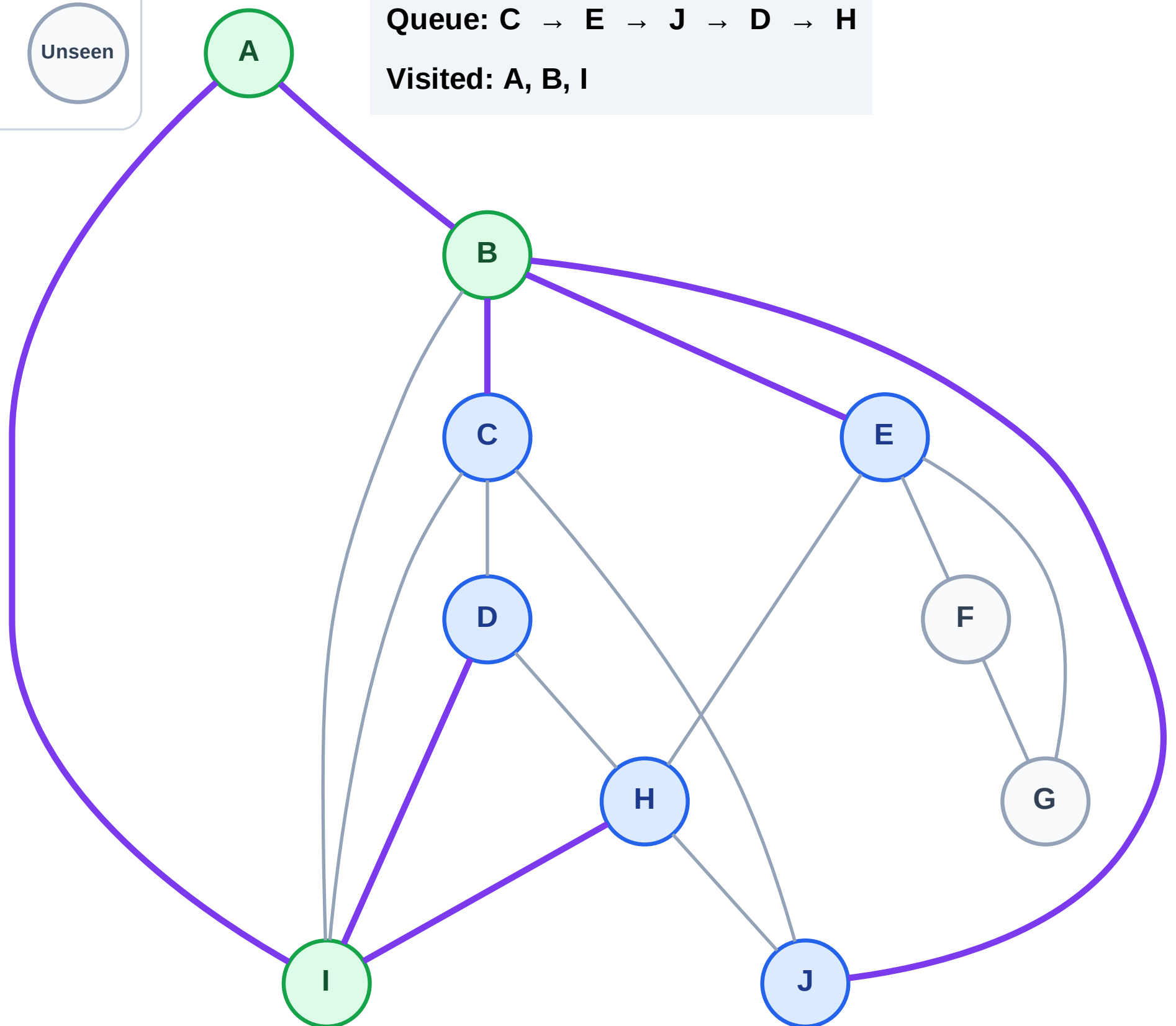
BFS Traversal

Step 7: Discover D, H from I



Queue: C → E → J → D → H

Visited: A, B, I



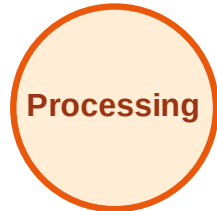
BFS Traversal

Step 8: Process node C

Legend



Complete



Processing



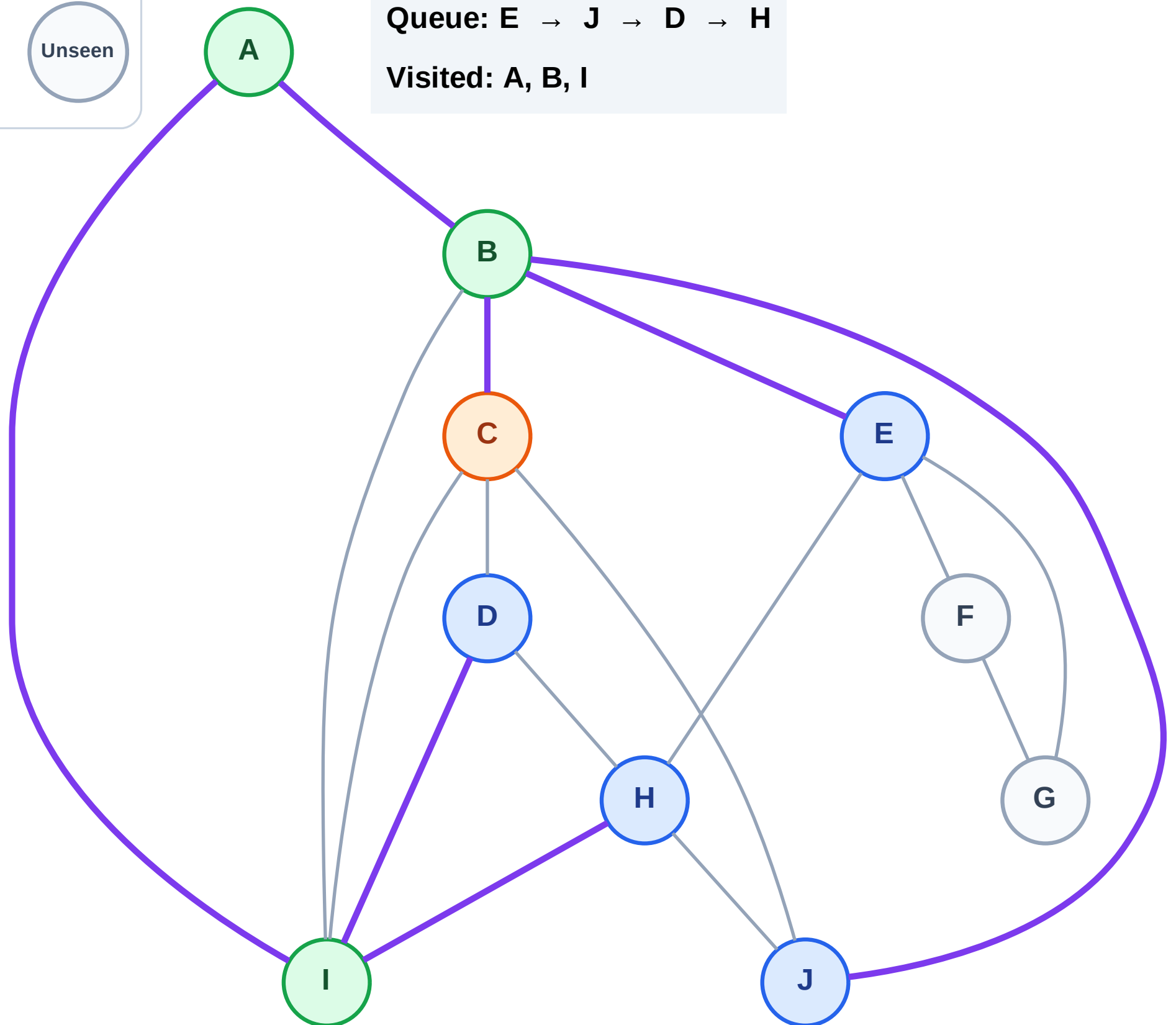
Discovered



Unseen

Queue: E → J → D → H

Visited: A, B, I



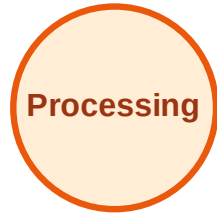
BFS Traversal

Step 9: No new neighbors from C

Legend



Complete



Processing



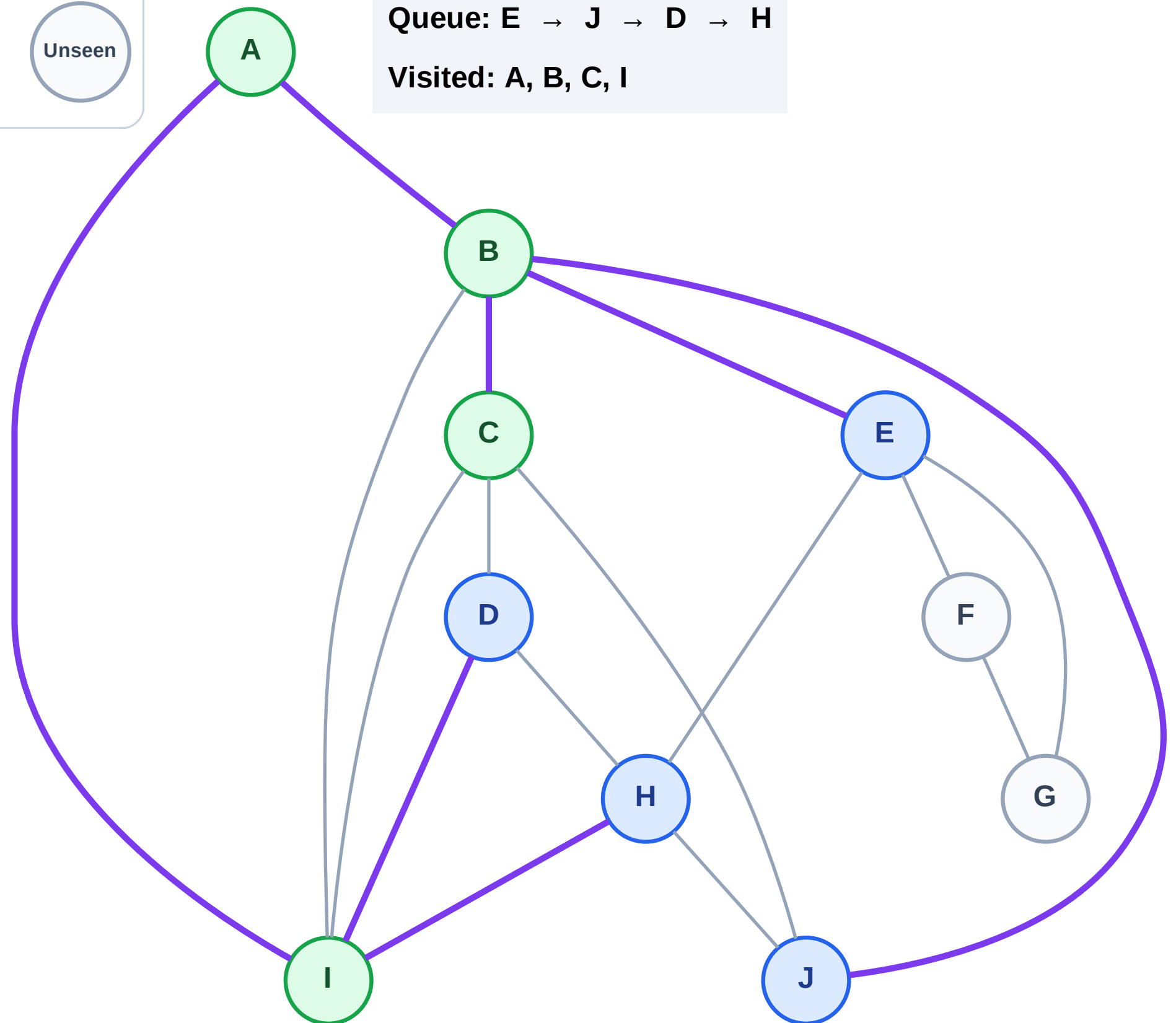
Discovered



Unseen

Queue: E → J → D → H

Visited: A, B, C, I



BFS Traversal

Step 10: Process node E

Legend

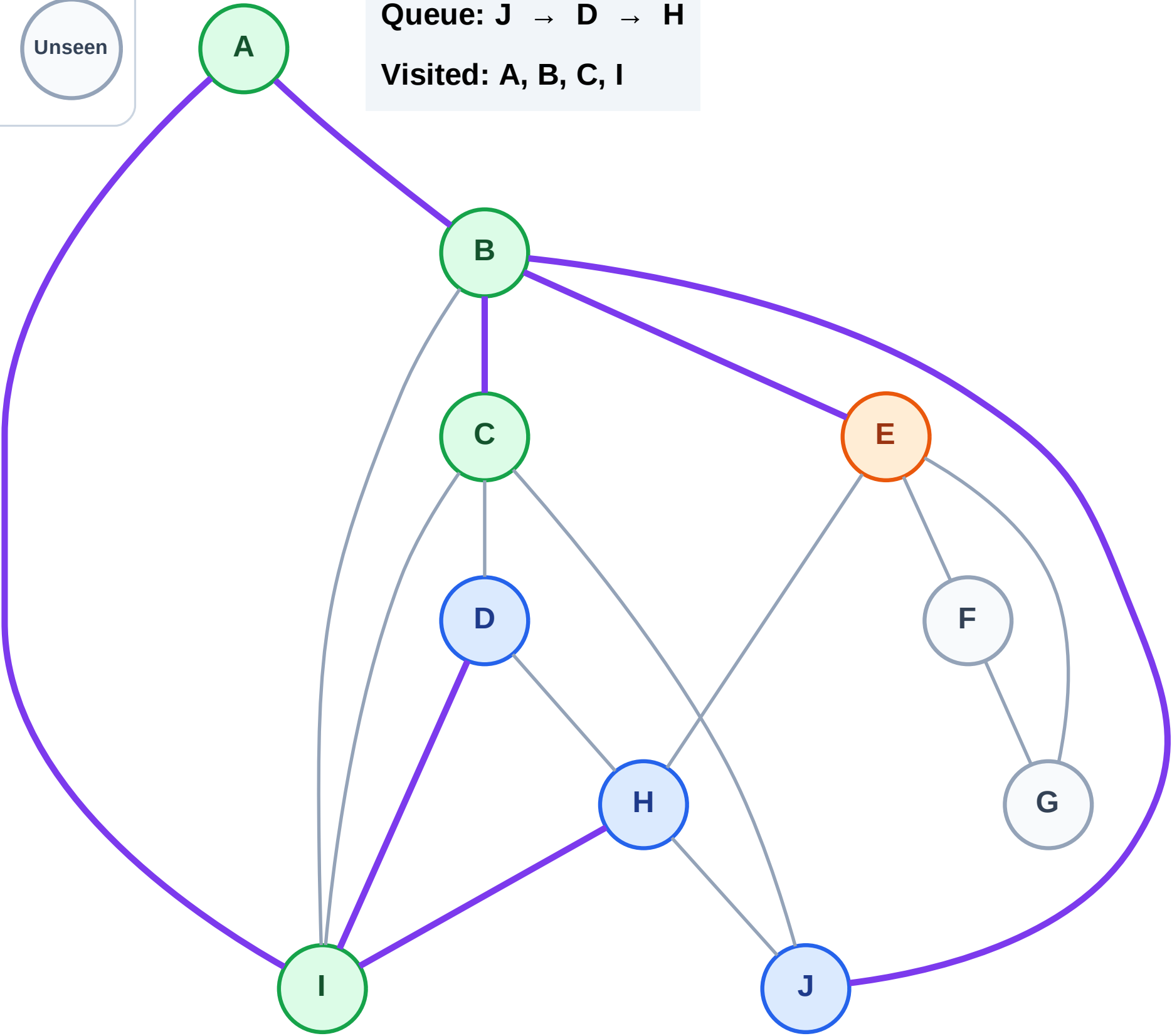
Complete

Processing

Discovered

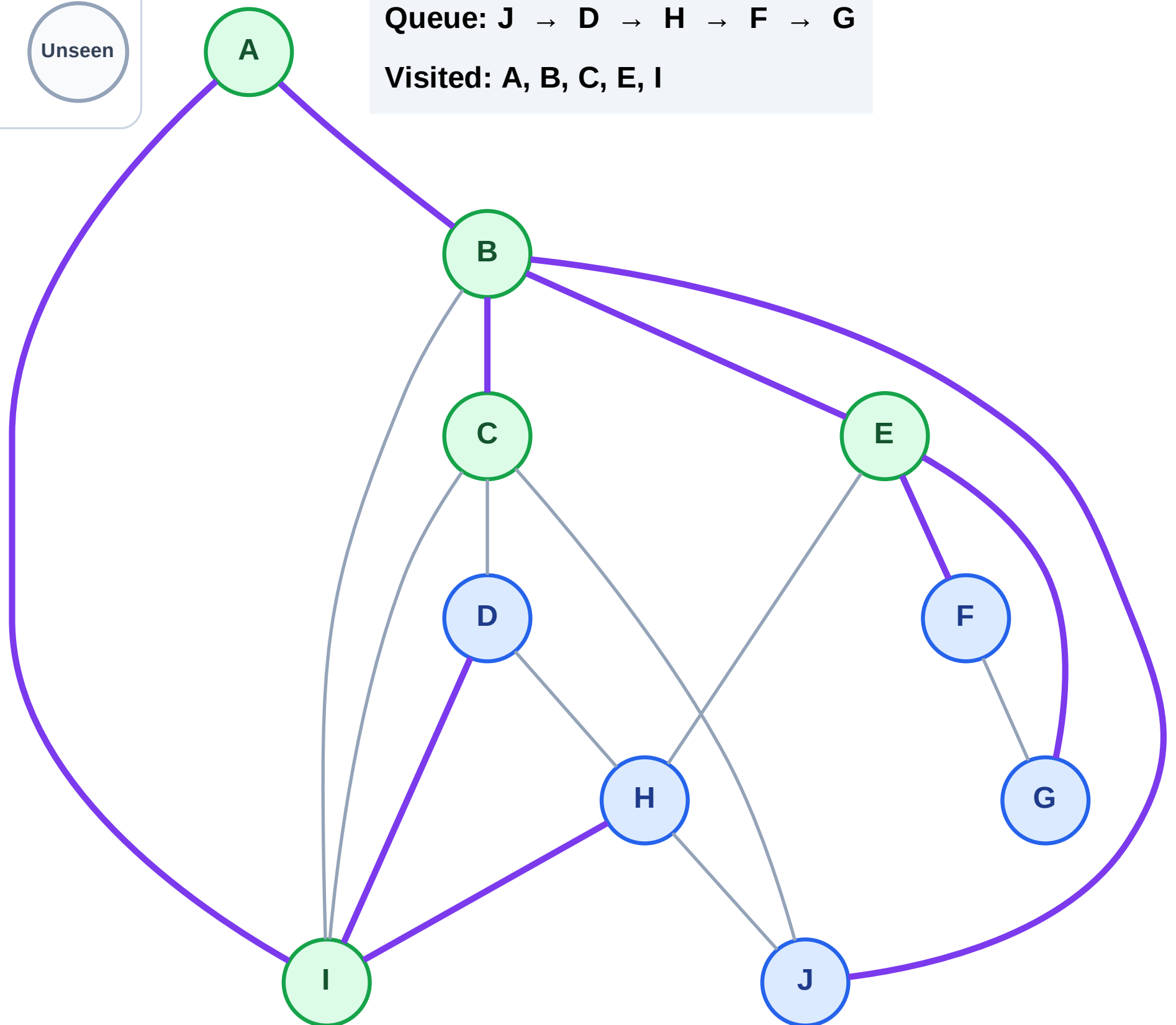
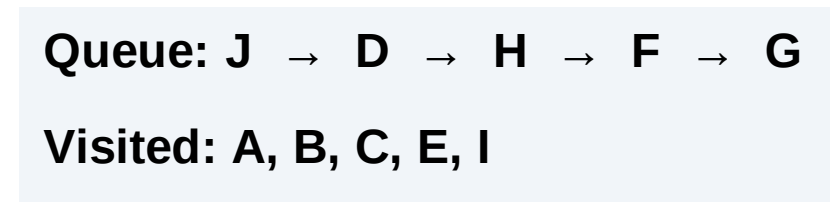
Unseen

Queue: J → D → H
Visited: A, B, C, I



Step 11: Discover F, G from E

Step 11: Discover F, G from E



BFS Traversal

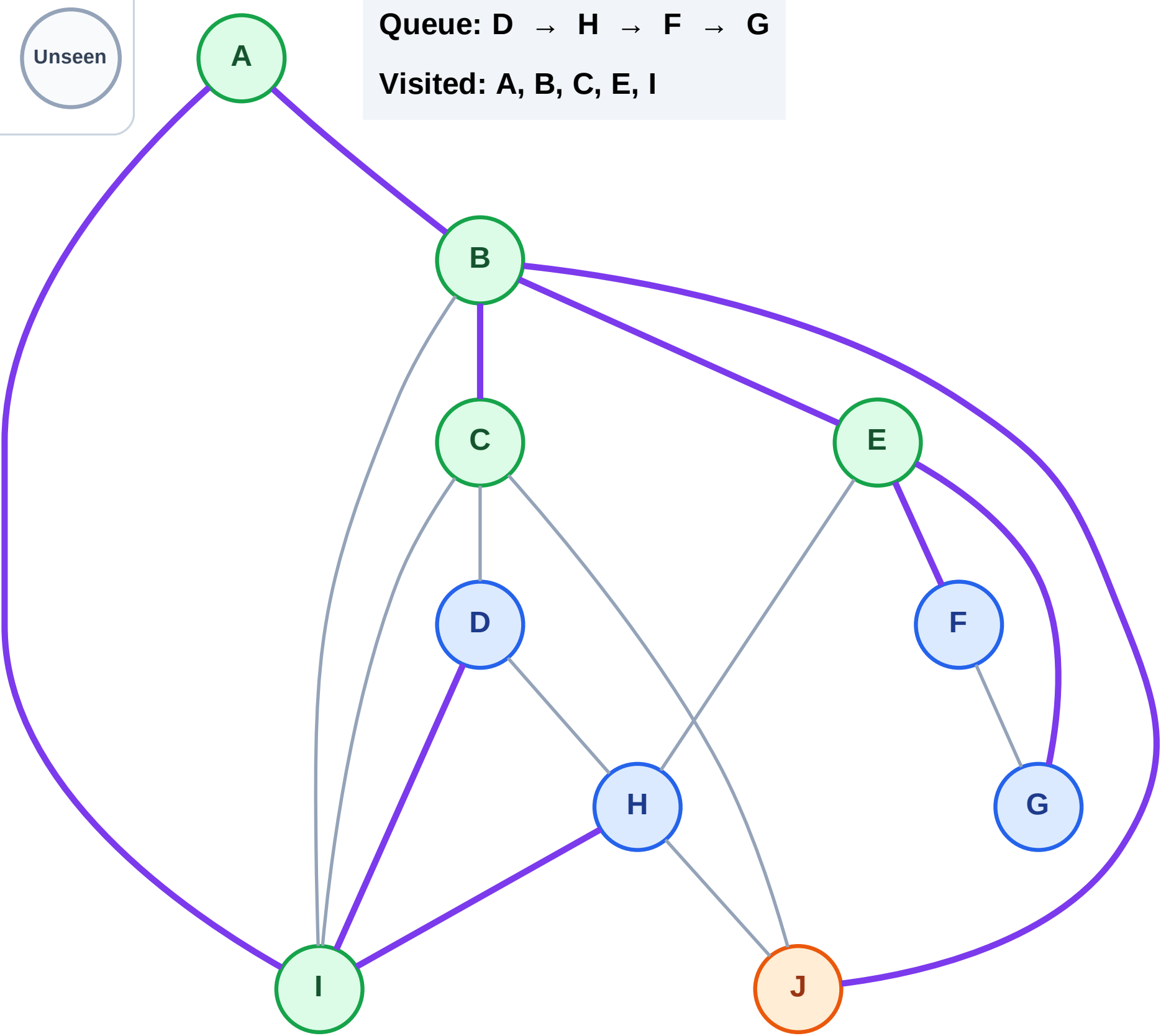
Step 12: Process node J

Legend



Queue: D → H → F → G

Visited: A, B, C, E, I



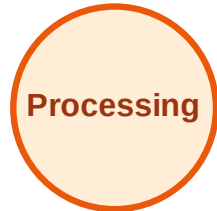
BFS Traversal

Step 13: No new neighbors from J

Legend



Complete



Processing



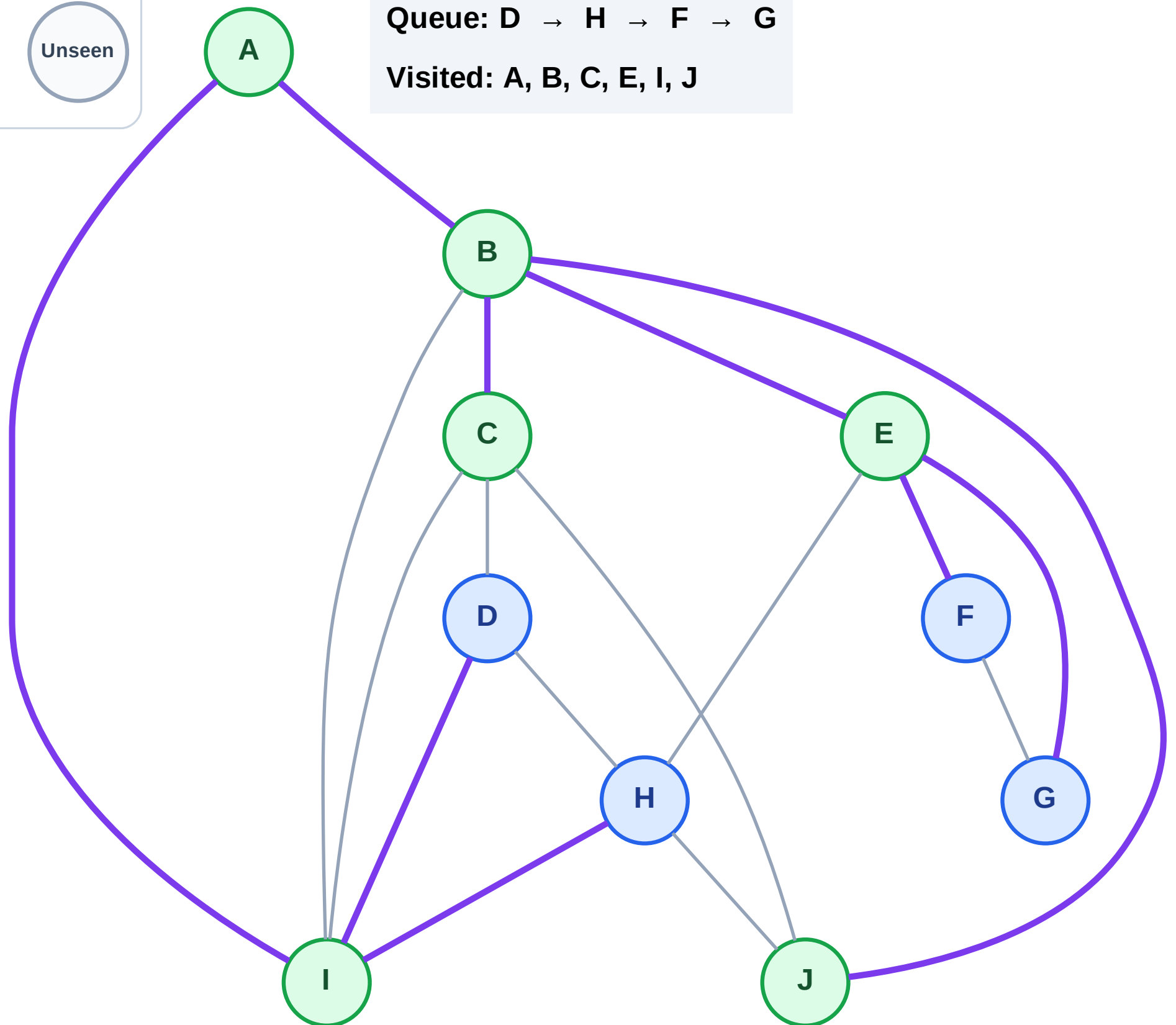
Discovered



Unseen

Queue: D → H → F → G

Visited: A, B, C, E, I, J



BFS Traversal

Step 14: Process node D

Legend

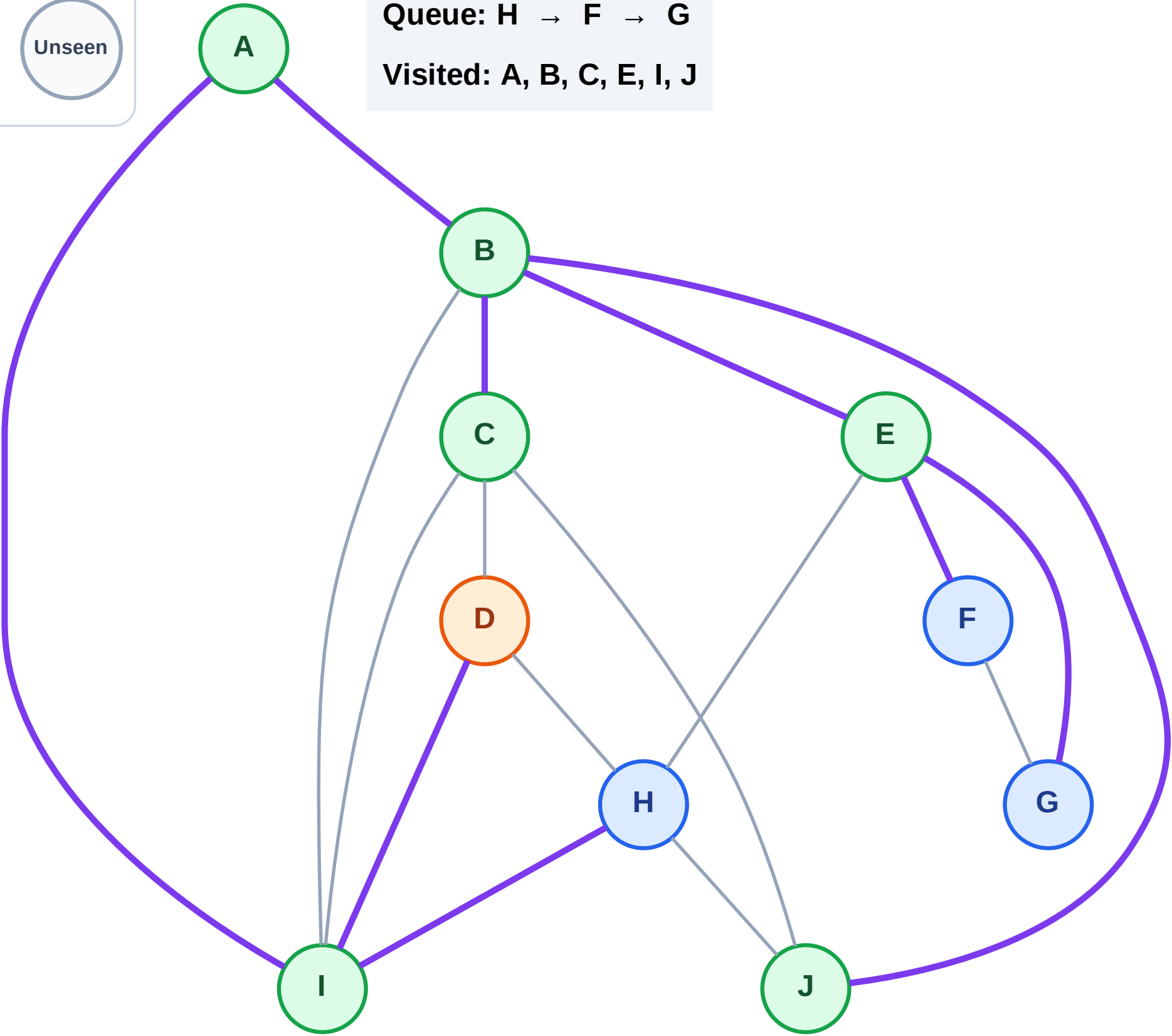
Complete

Processing

Discovered

Unseen

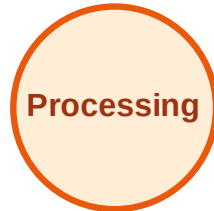
Queue: H → F → G
Visited: A, B, C, E, I, J



BFS Traversal

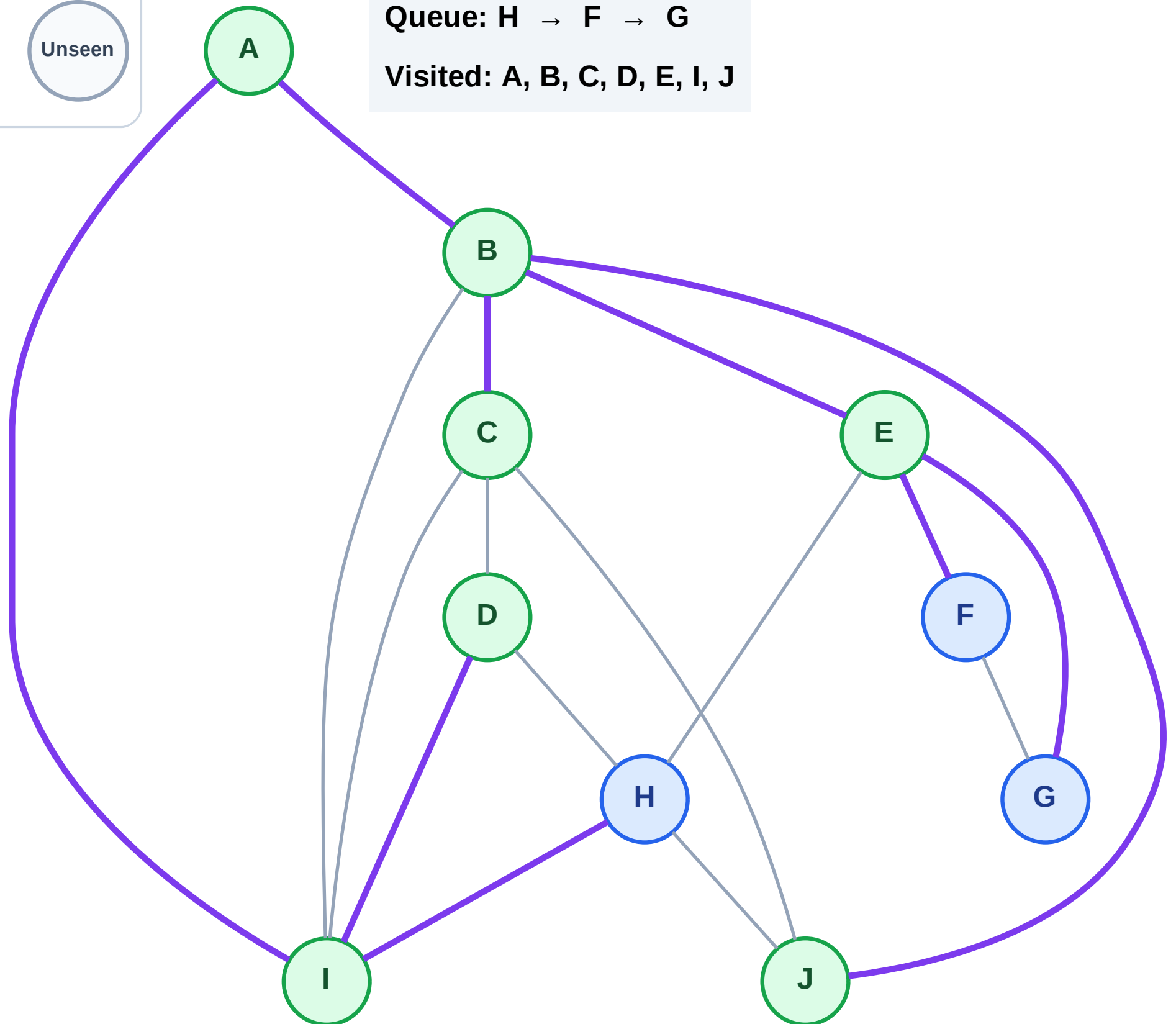
Step 15: No new neighbors from D

Legend



Queue: H → F → G

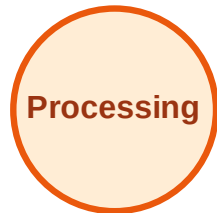
Visited: A, B, C, D, E, I, J



BFS Traversal

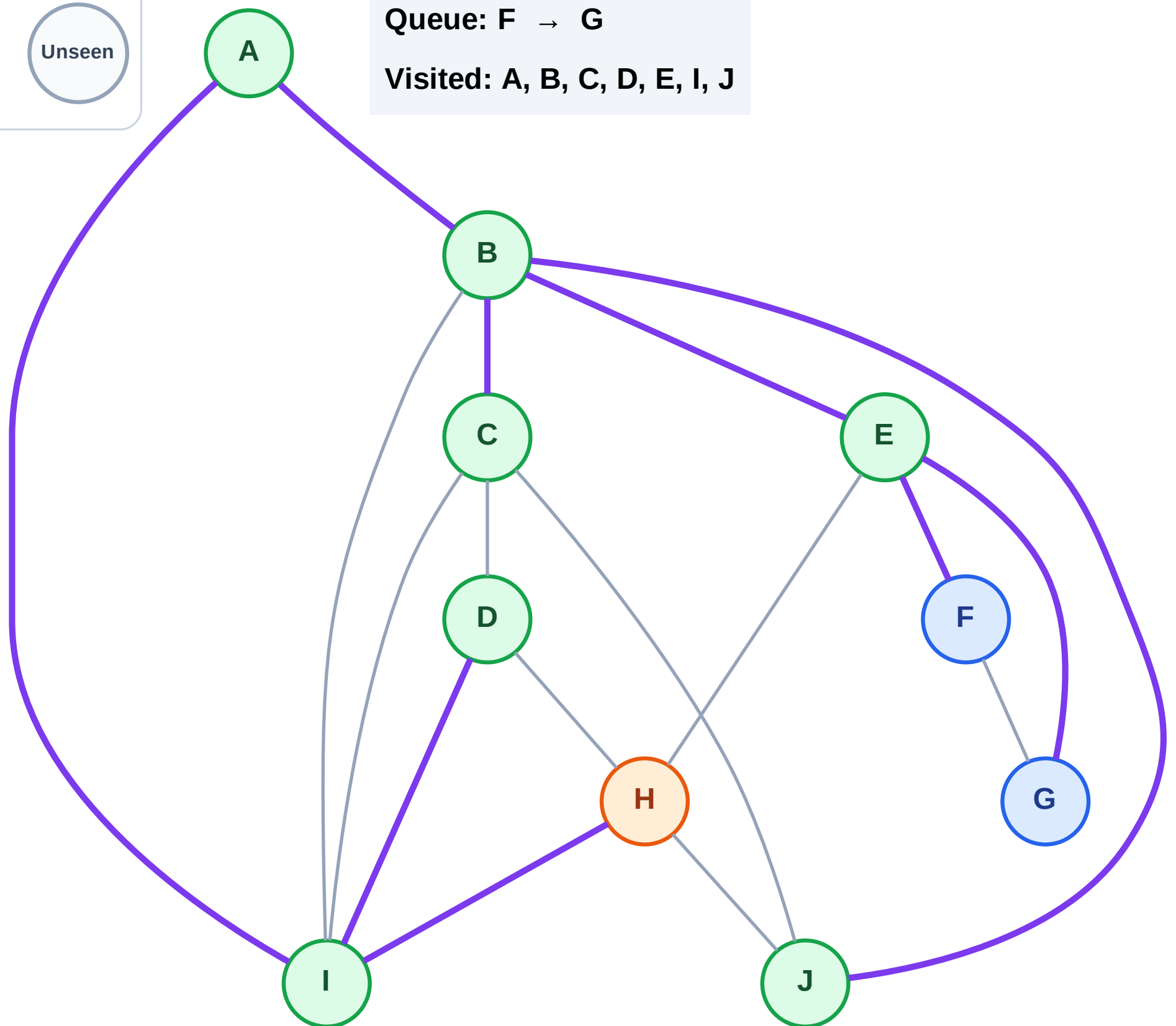
Step 16: Process node H

Legend



Queue: F → G

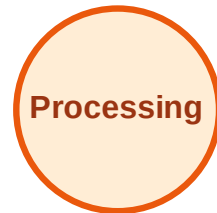
Visited: A, B, C, D, E, I, J



BFS Traversal

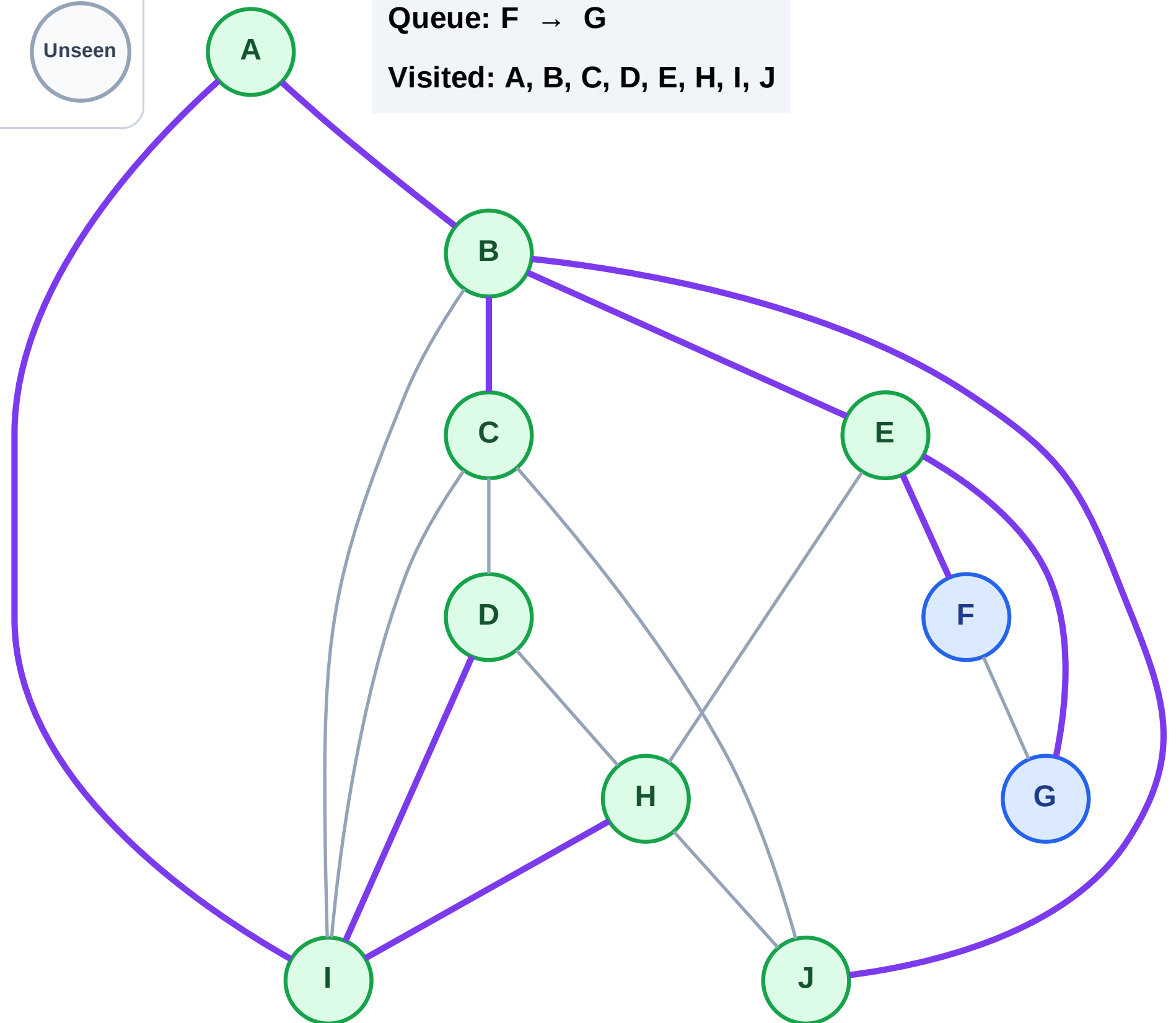
Step 17: No new neighbors from H

Legend



Queue: F → G

Visited: A, B, C, D, E, H, I, J



BFS Traversal

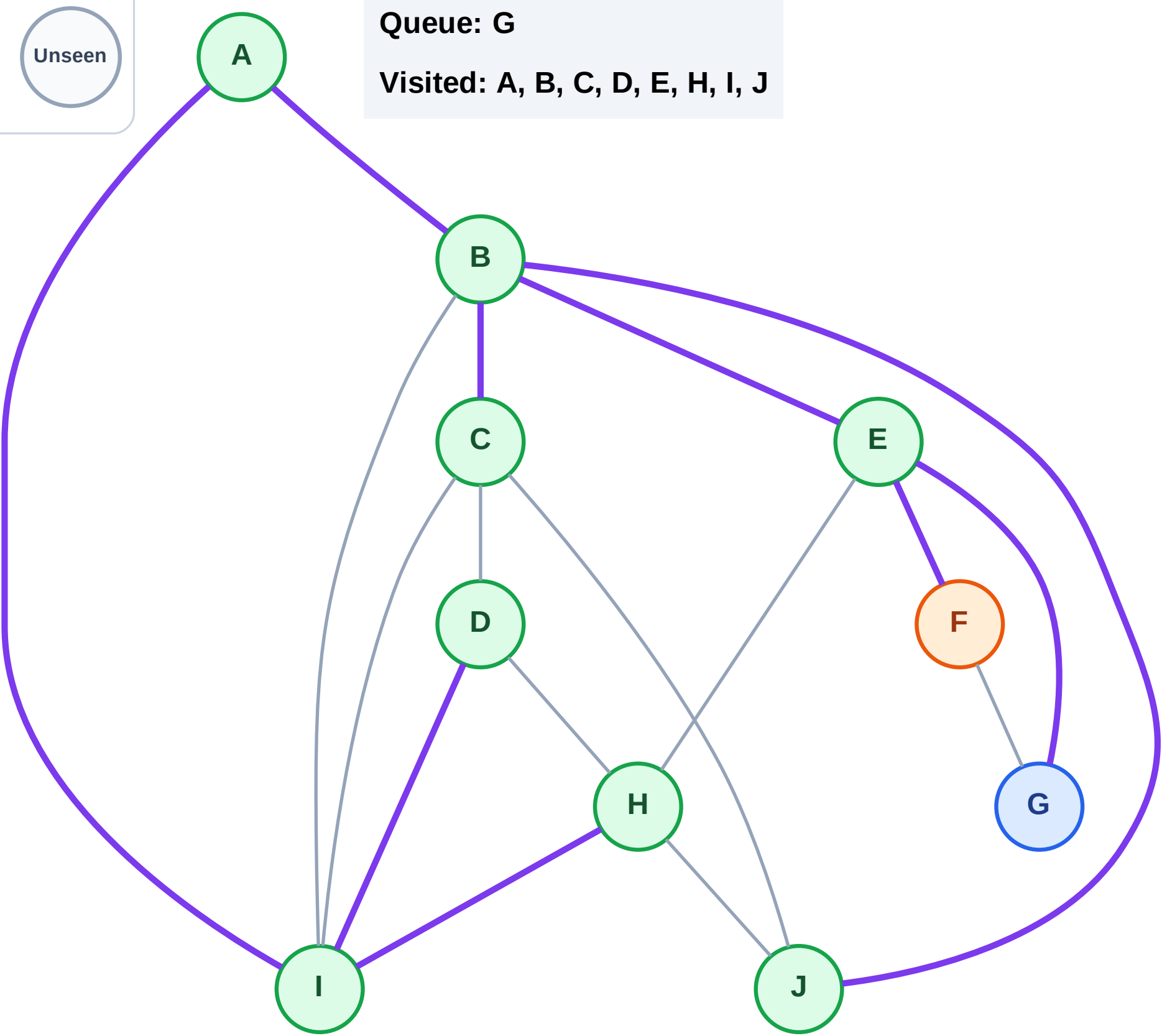
Step 18: Process node F

Legend



Queue: G

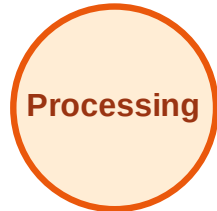
Visited: A, B, C, D, E, H, I, J



BFS Traversal

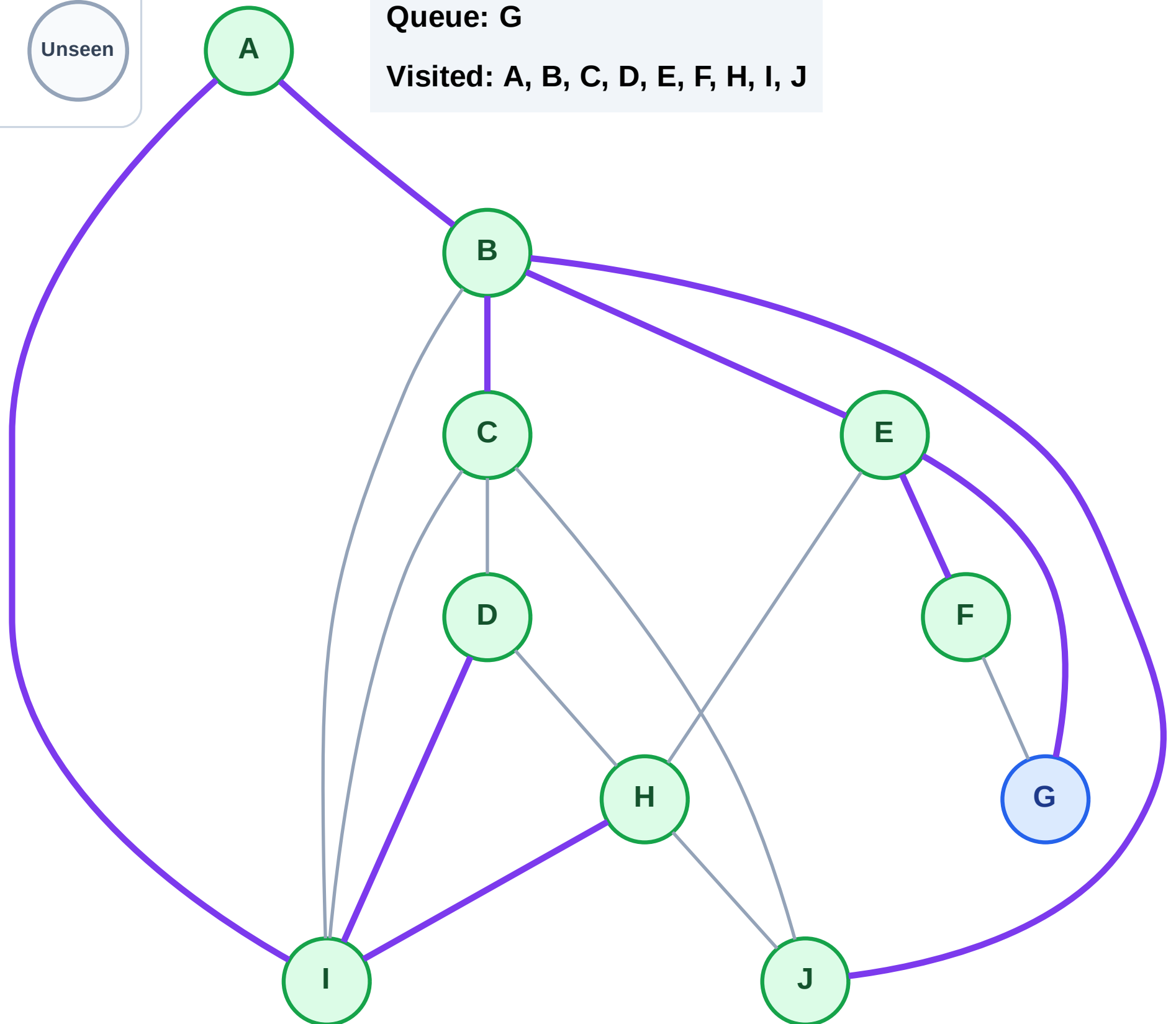
Step 19: No new neighbors from F

Legend



Queue: G

Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

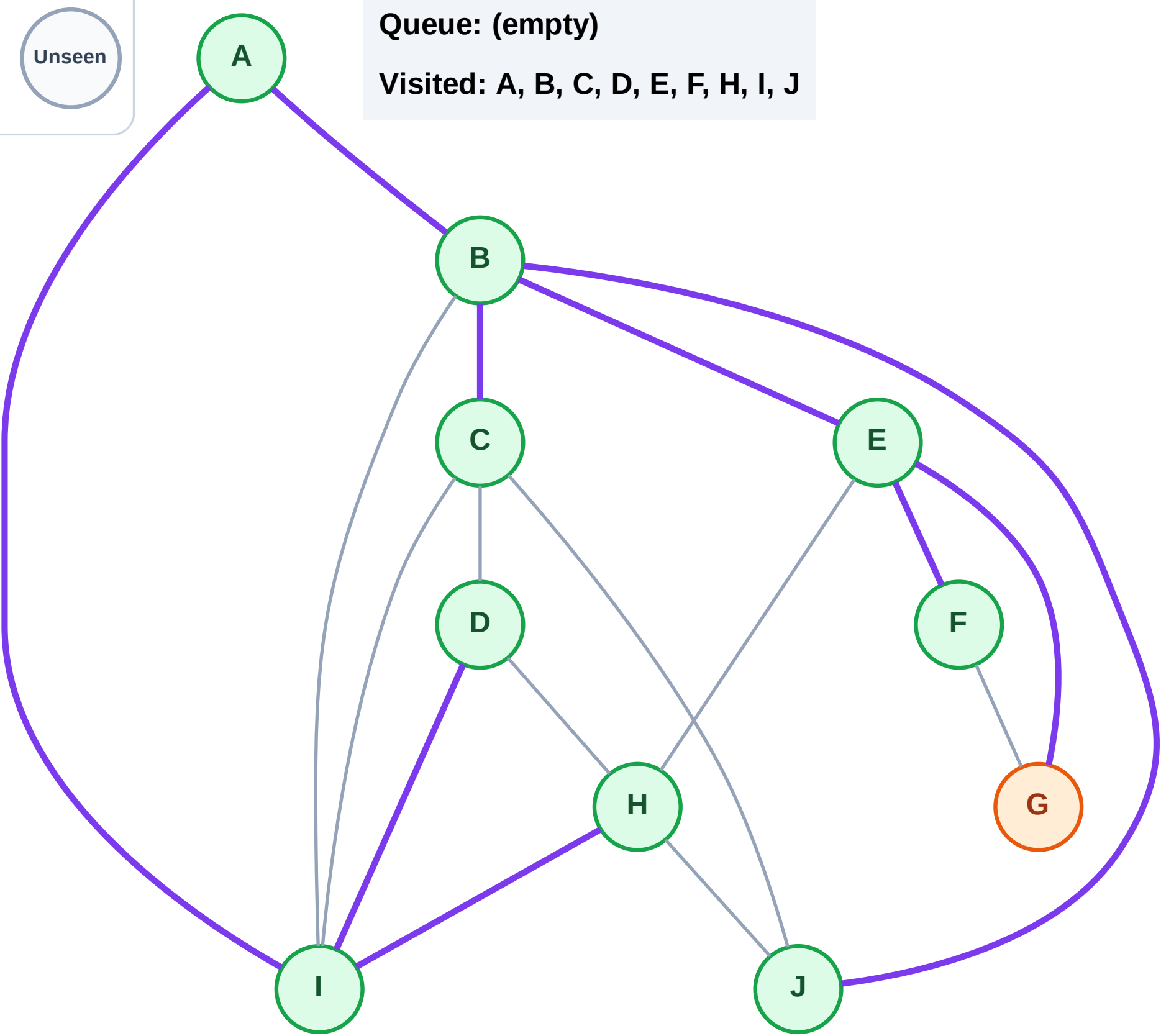
Step 20: Process node G

Legend



Queue: (empty)

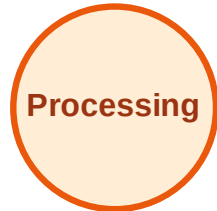
Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

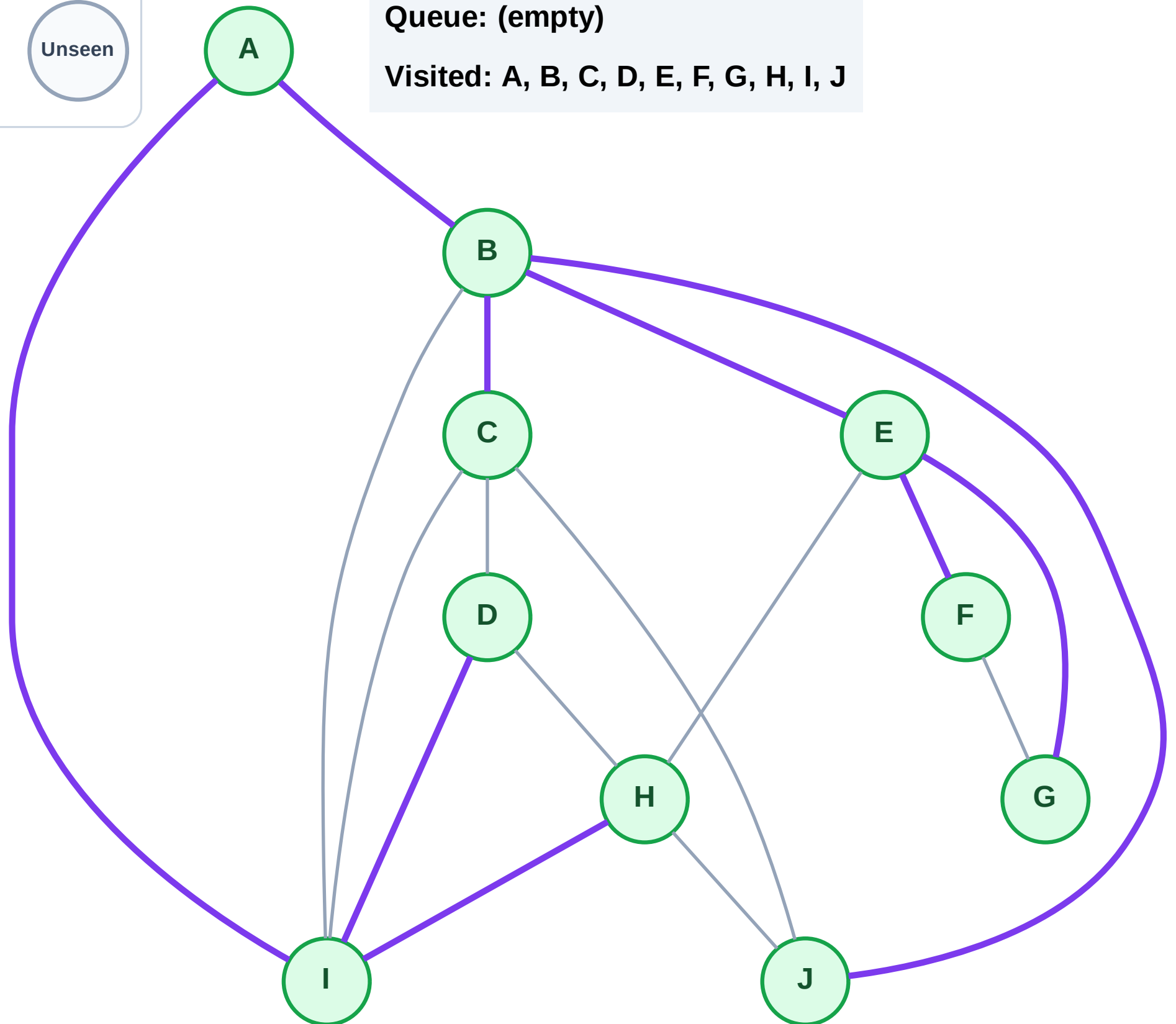
Step 21: No new neighbors from G

Legend



Queue: (empty)

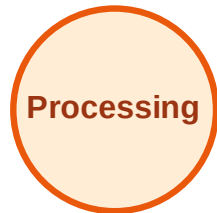
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

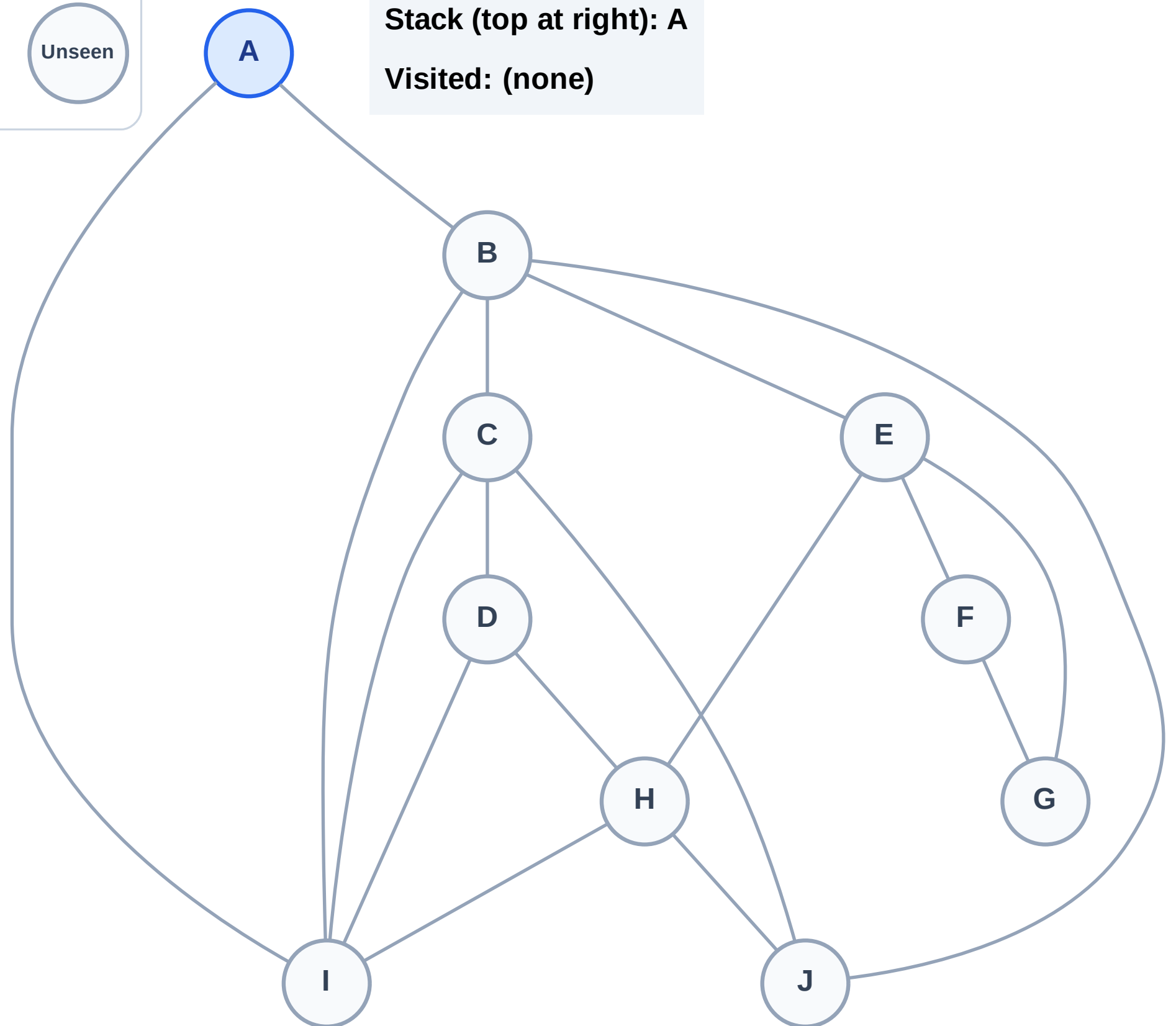
Step 1: Start at A

Legend



Stack (top at right): A

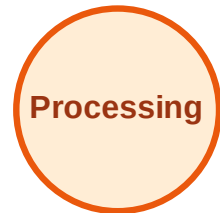
Visited: (none)



DFS Traversal

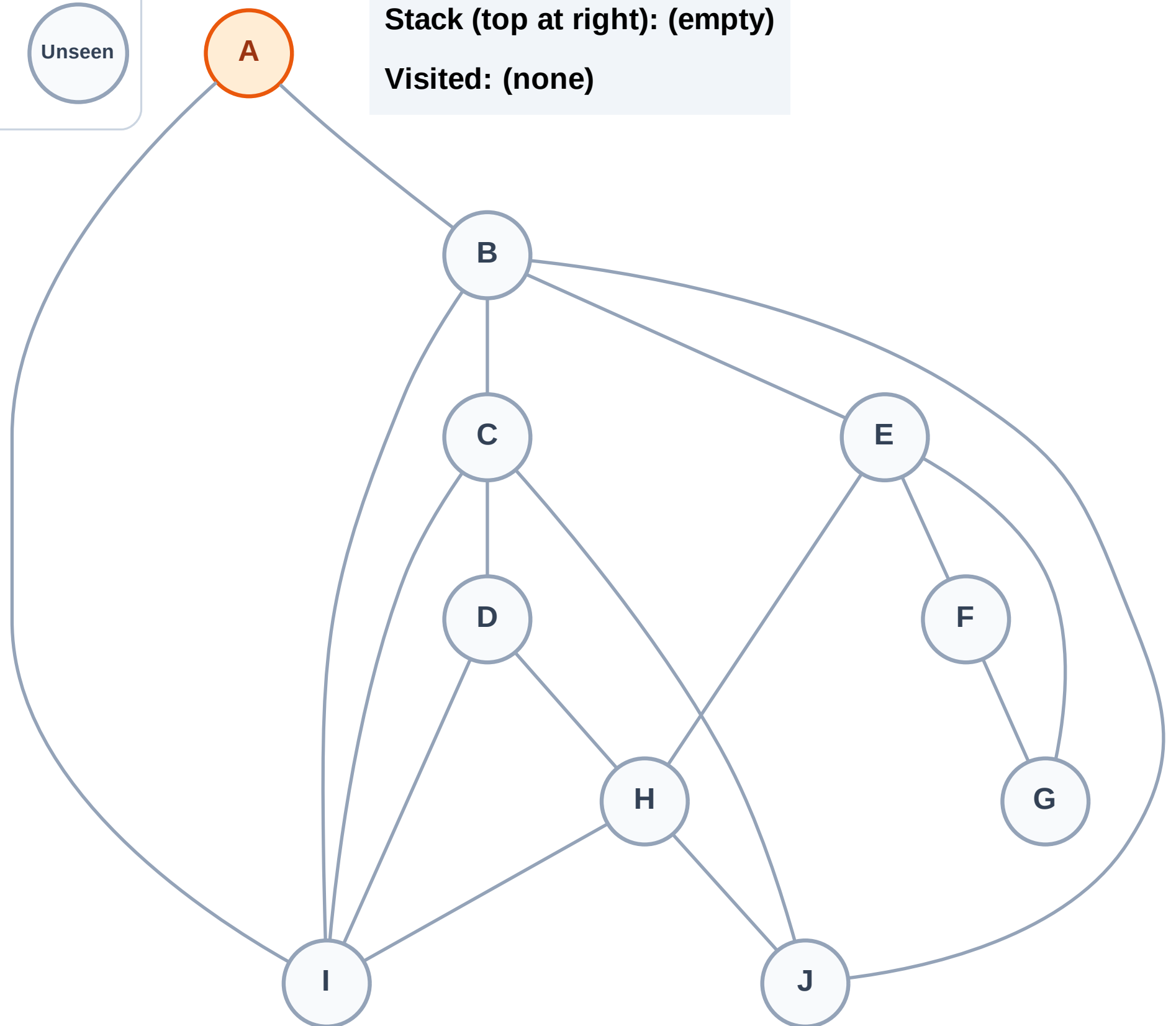
Step 2: Process node A

Legend



Stack (top at right): (empty)

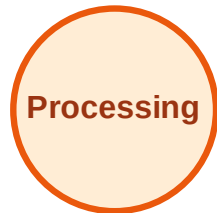
Visited: (none)



DFS Traversal

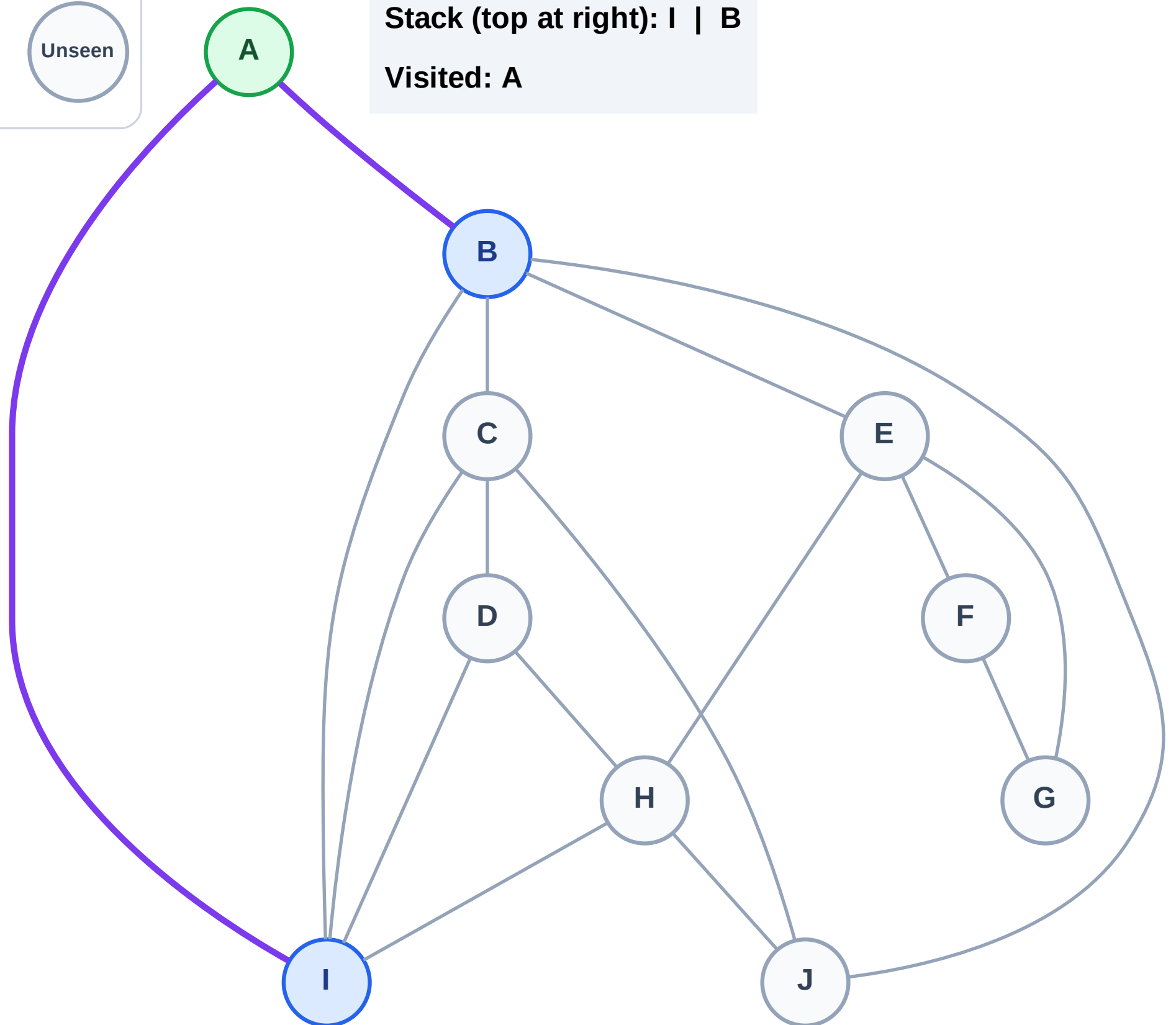
Step 3: Discover I, B from A

Legend



Stack (top at right): I | B

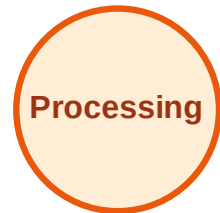
Visited: A



DFS Traversal

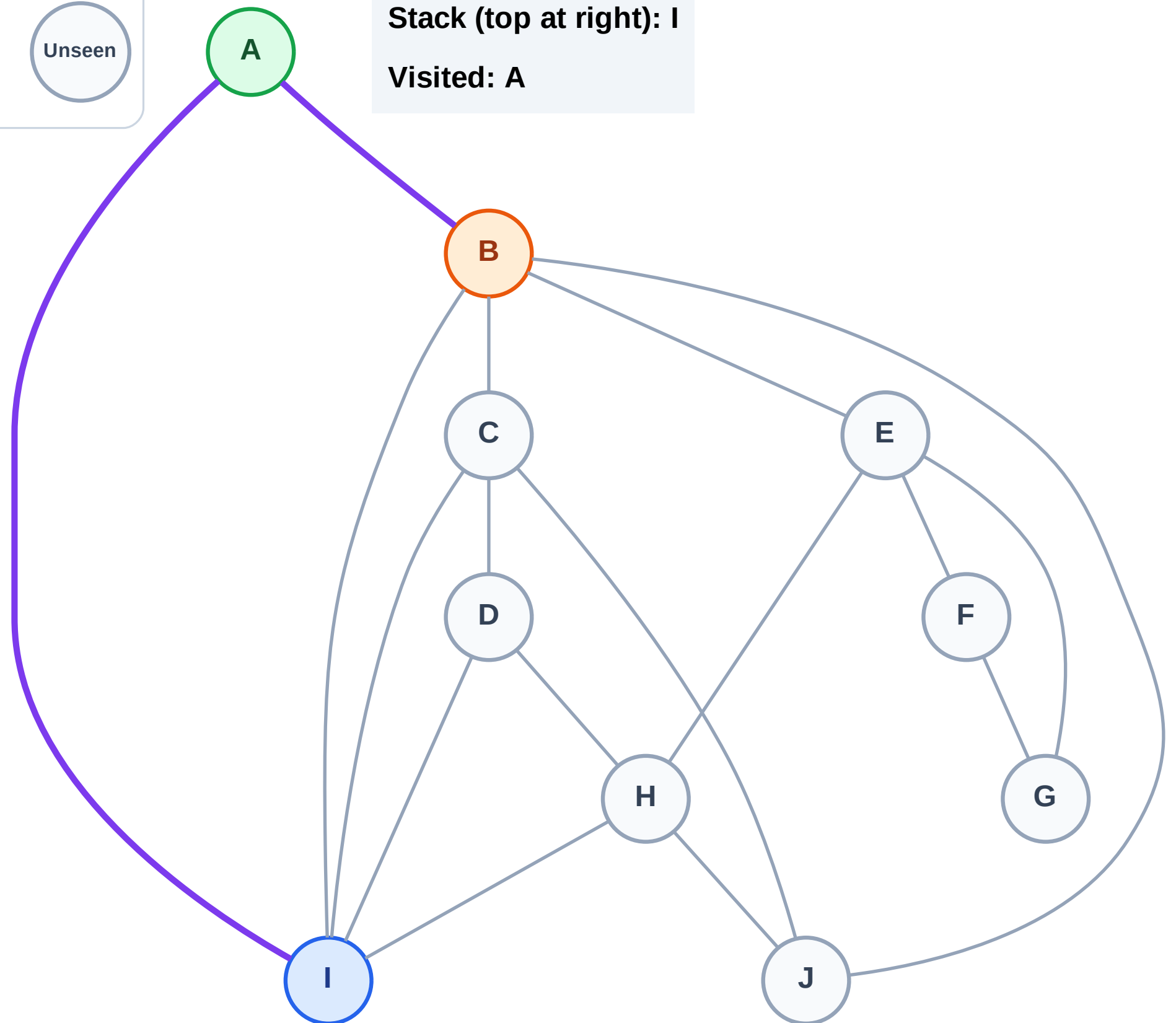
Step 4: Process node B

Legend



Stack (top at right): I

Visited: A



Step 5: Discover J, E, C from B

Stack (top at right): I | J | E | C

Legend

Processing

Unseen



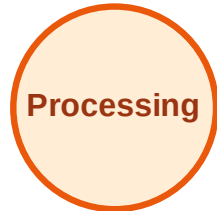
DFS Traversal

Step 6: Process node C

Legend



Complete



Processing



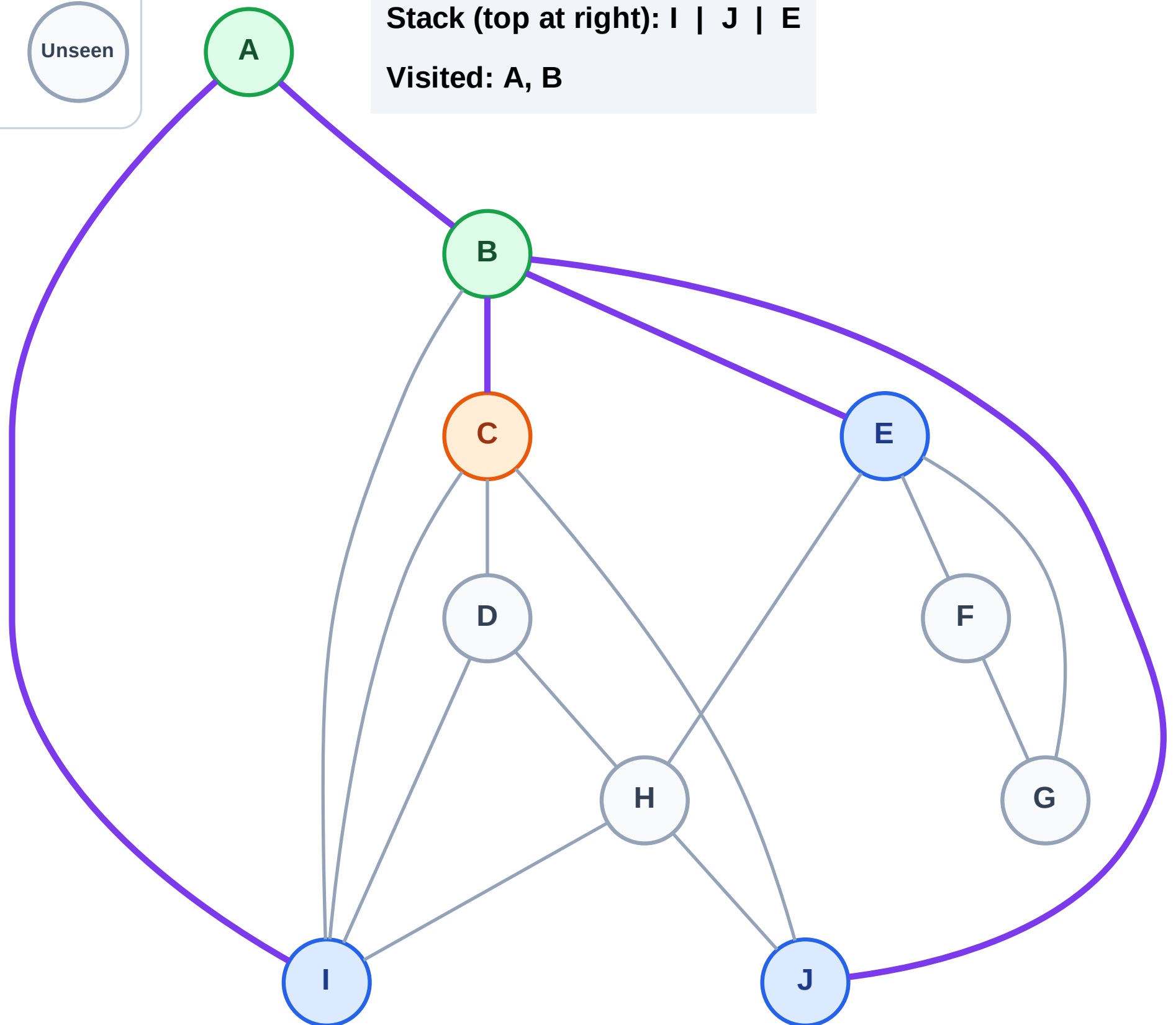
Discovered



Unseen

Stack (top at right): I | J | E

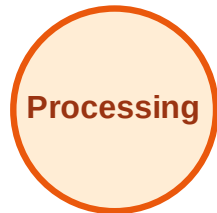
Visited: A, B



DFS Traversal

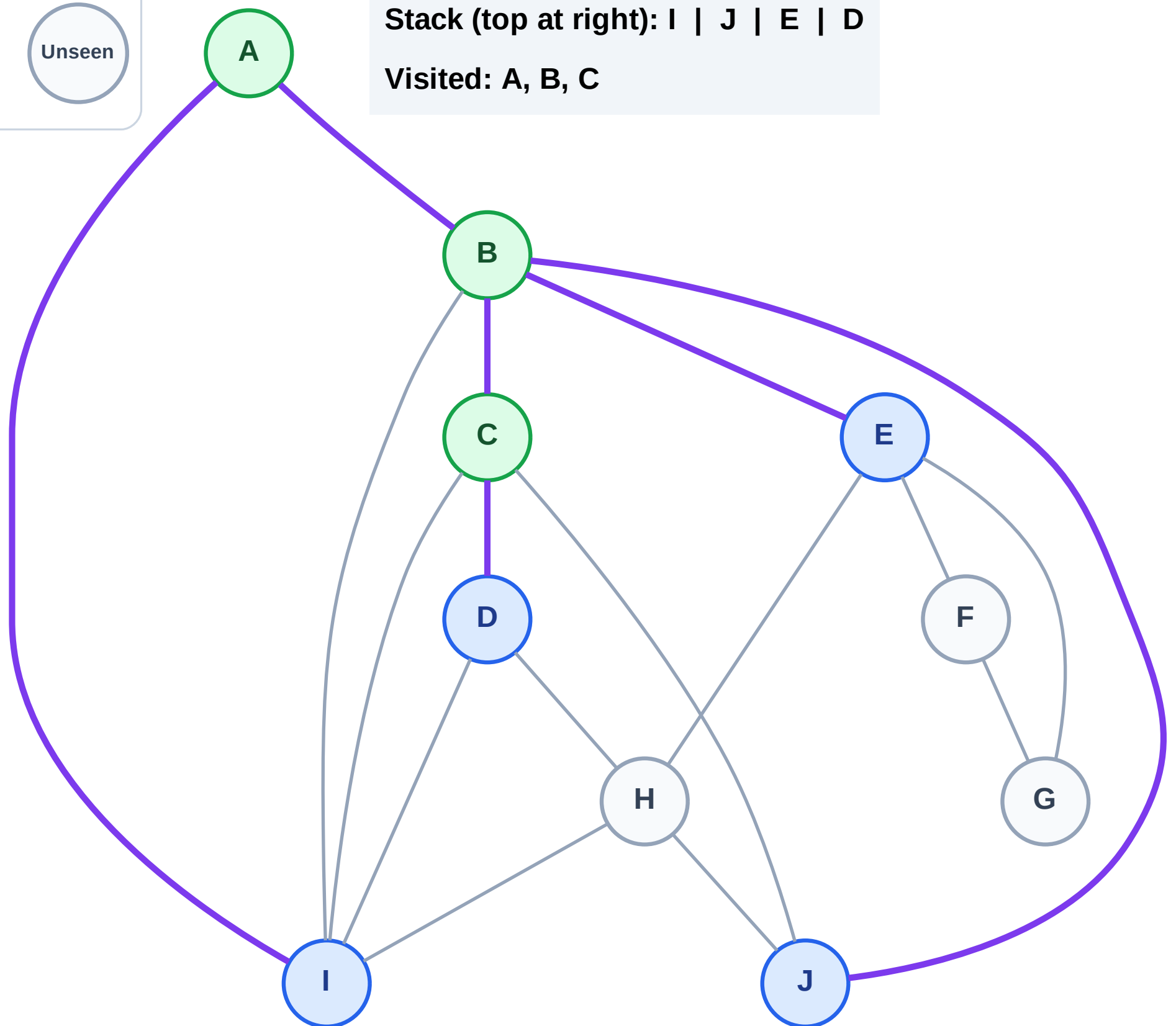
Step 7: Discover D from C

Legend

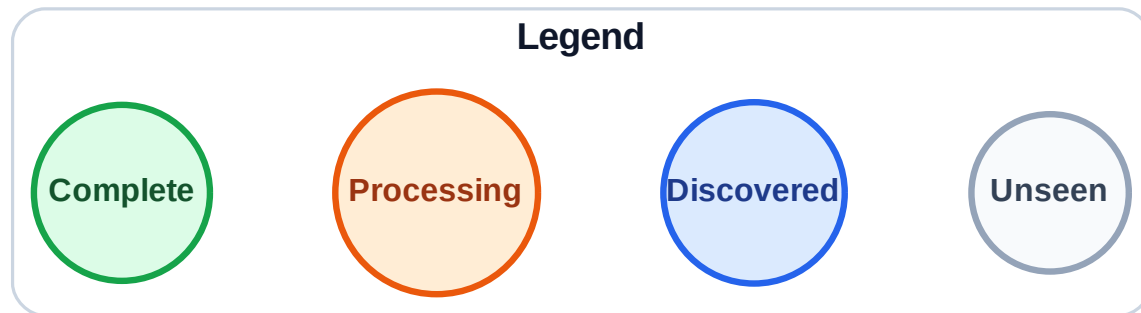


Stack (top at right): I | J | E | D

Visited: A, B, C

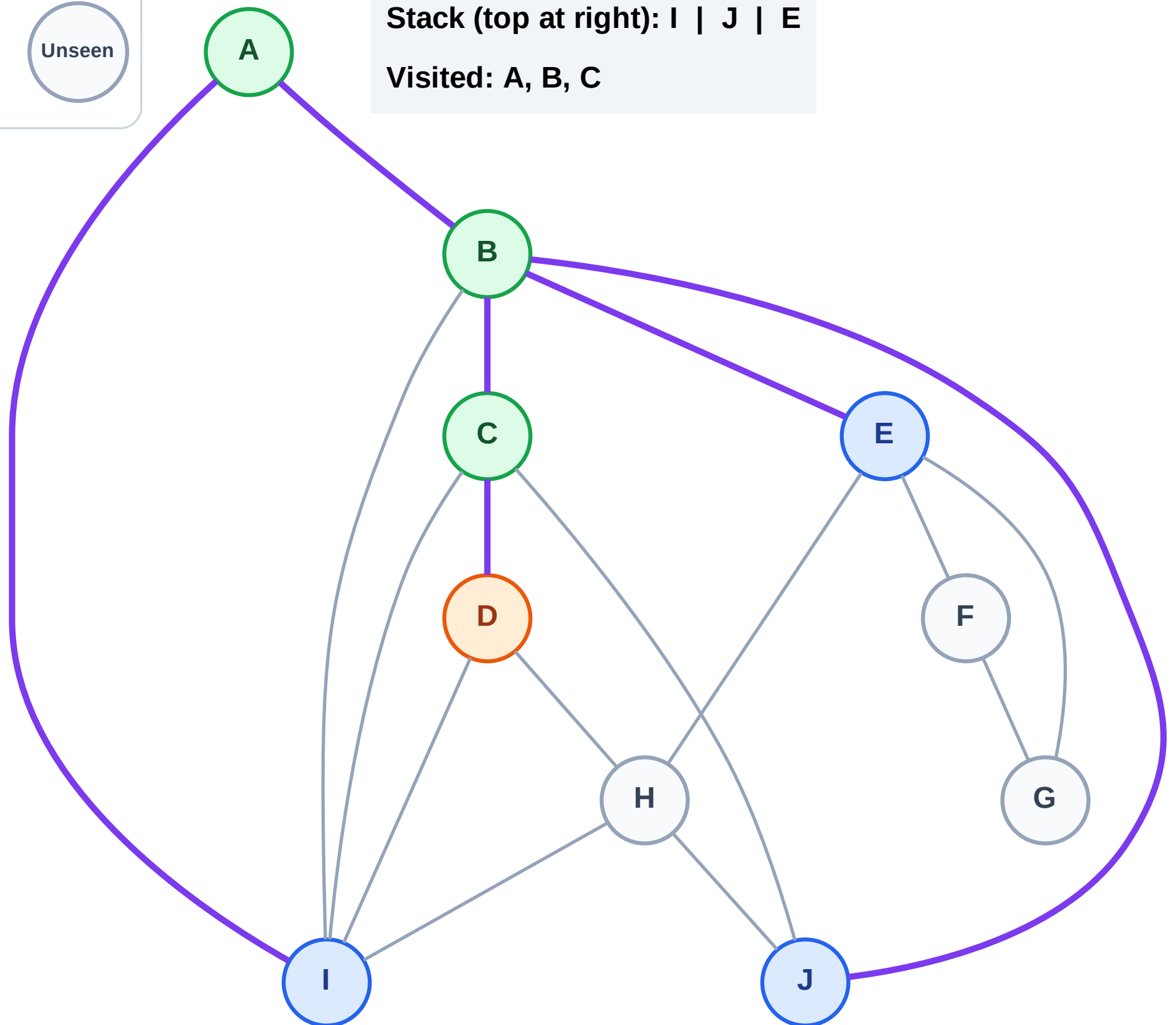


Step 8: Process node D



Stack (top at right): I | J | E

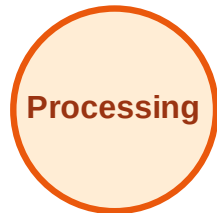
Visited: A, B, C



DFS Traversal

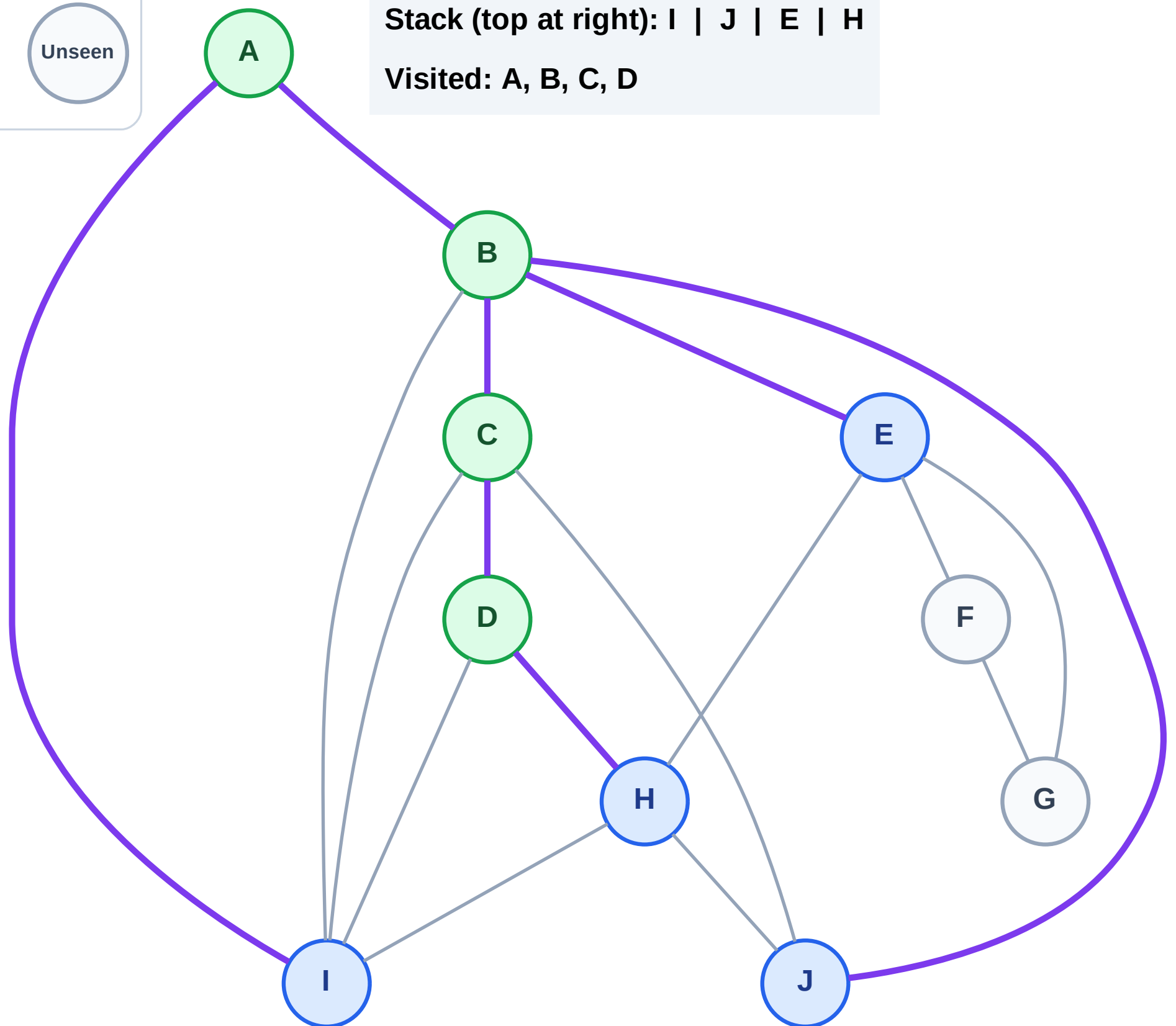
Step 9: Discover H from D

Legend



Stack (top at right): I | J | E | H

Visited: A, B, C, D



DFS Traversal

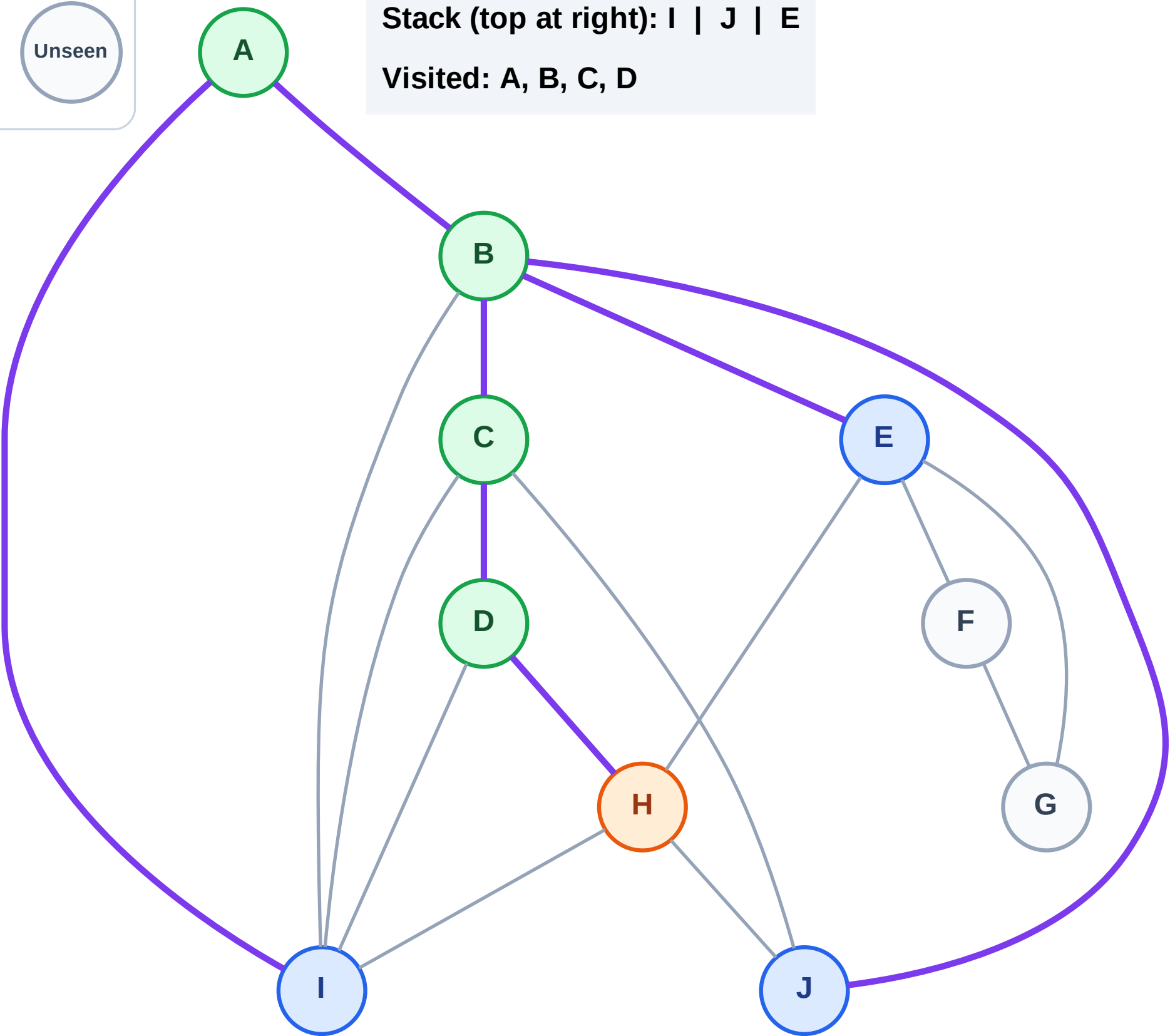
Step 10: Process node H

Legend



Stack (top at right): I | J | E

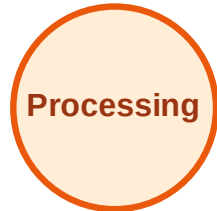
Visited: A, B, C, D



DFS Traversal

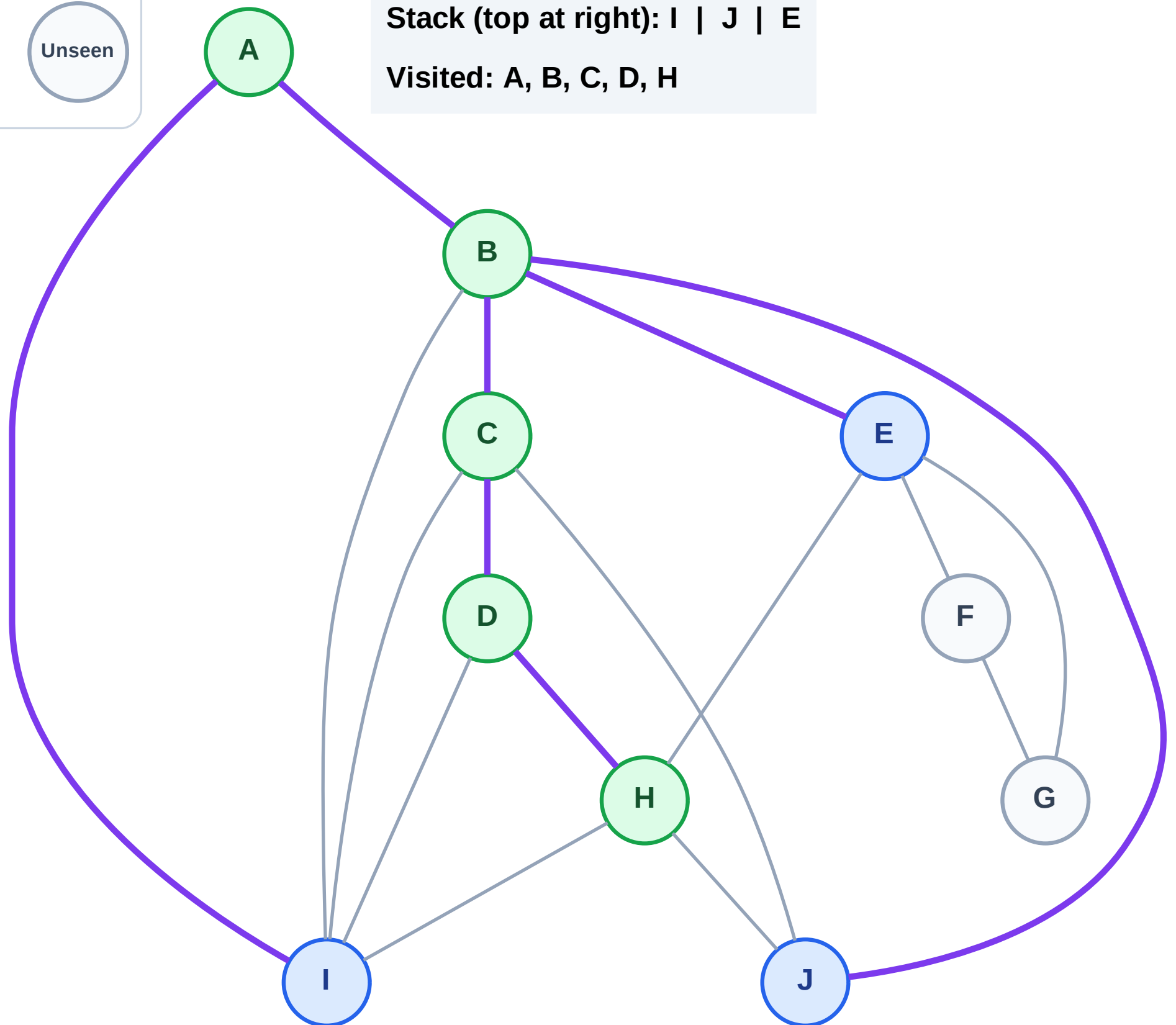
Step 11: No new neighbors from H

Legend



Stack (top at right): I | J | E

Visited: A, B, C, D, H



DFS Traversal

Step 12: Process node E

Legend

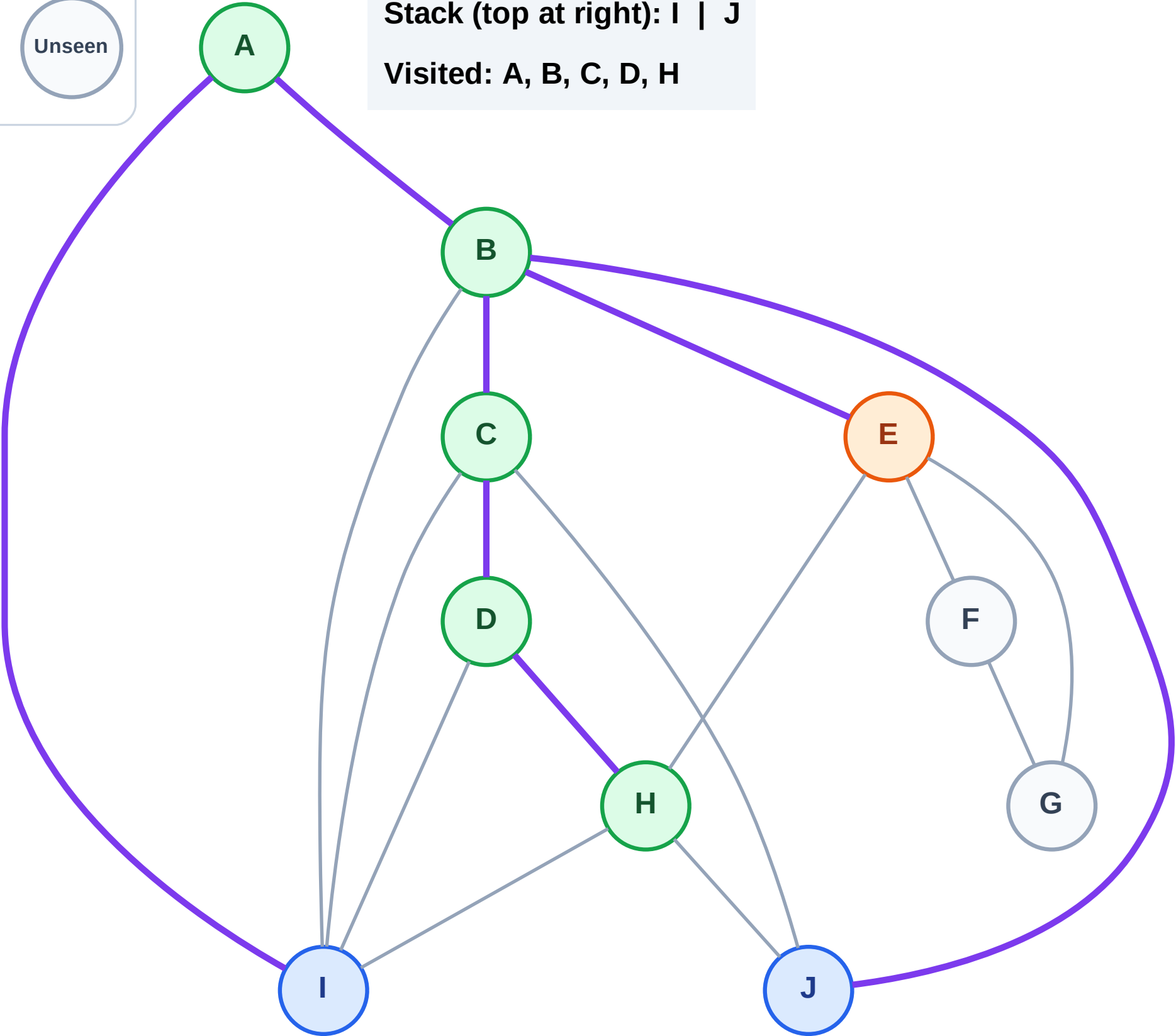
Complete

Processing

Discovered

Unseen

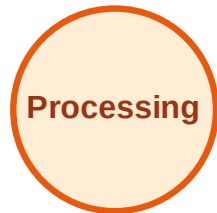
Stack (top at right): I | J
Visited: A, B, C, D, H



DFS Traversal

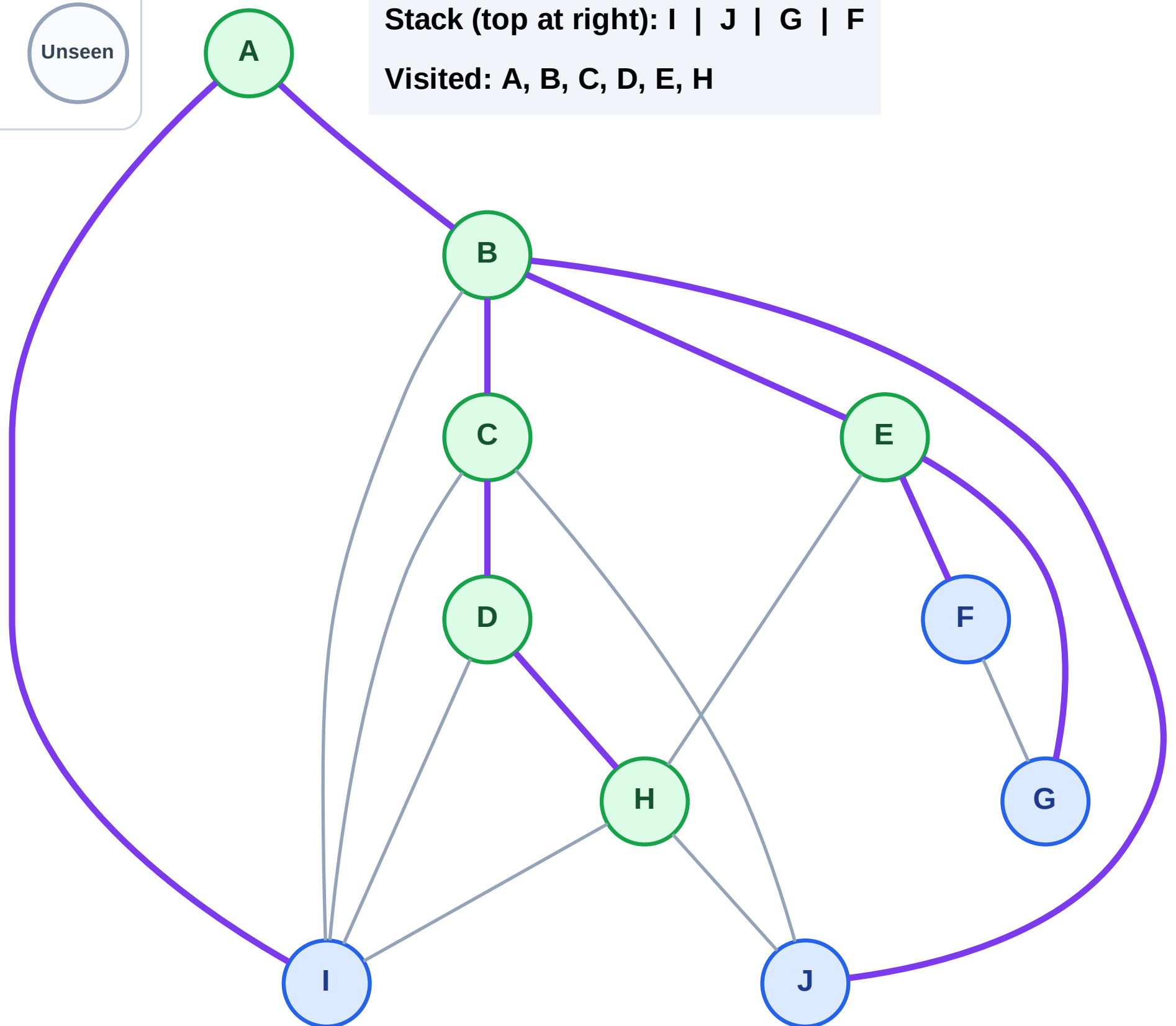
Step 13: Discover G, F from E

Legend



Stack (top at right): I | J | G | F

Visited: A, B, C, D, E, H



DFS Traversal

Step 14: Process node F

Legend

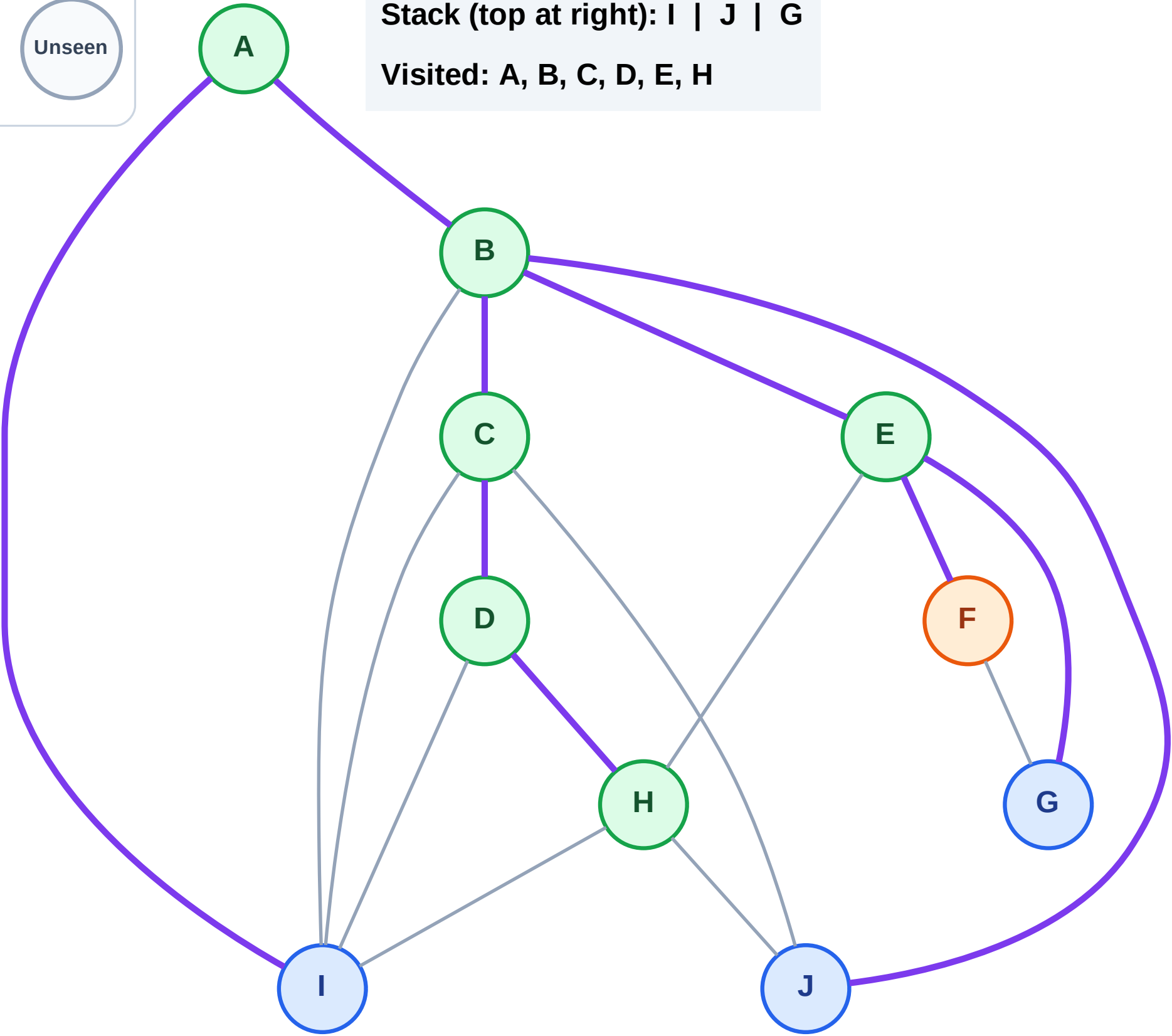
Complete

Processing

Discovered

Unseen

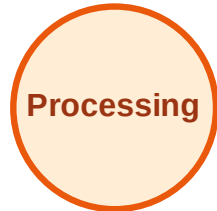
Stack (top at right): I | J | G
Visited: A, B, C, D, E, H



DFS Traversal

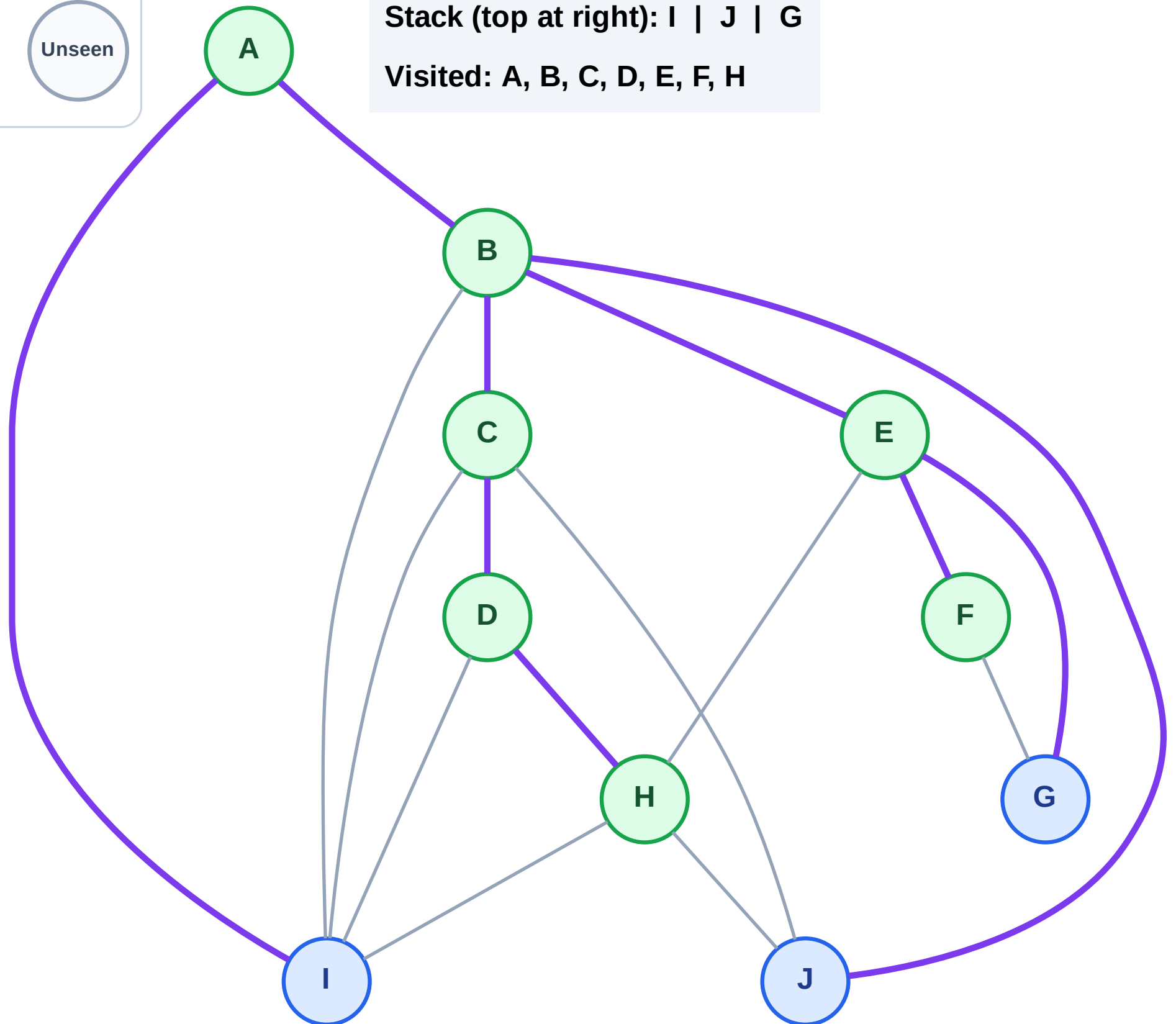
Step 15: No new neighbors from F

Legend



Stack (top at right): I | J | G

Visited: A, B, C, D, E, F, H



DFS Traversal

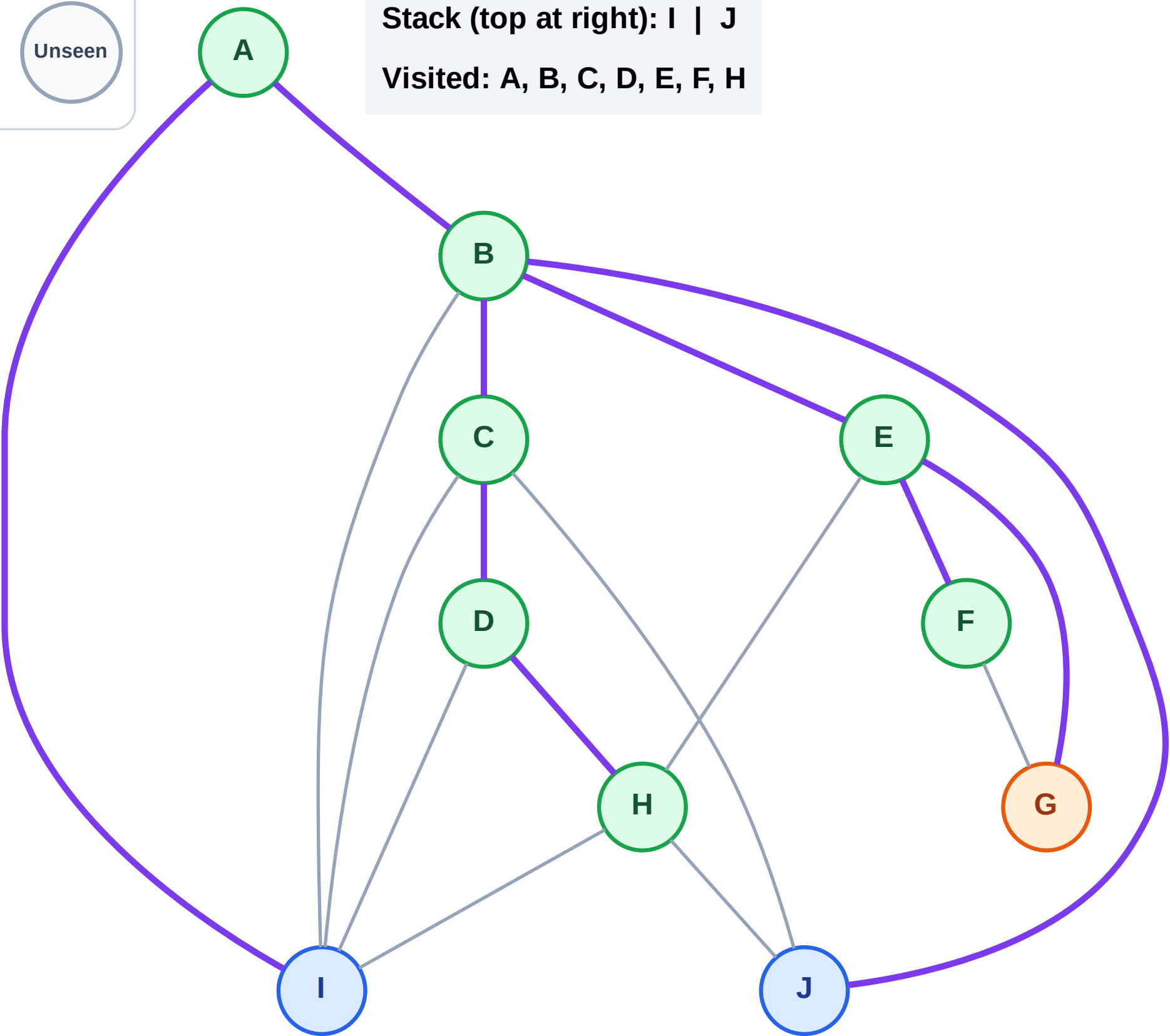
Step 16: Process node G

Legend



Stack (top at right): I | J

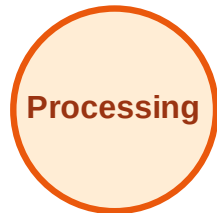
Visited: A, B, C, D, E, F, H



DFS Traversal

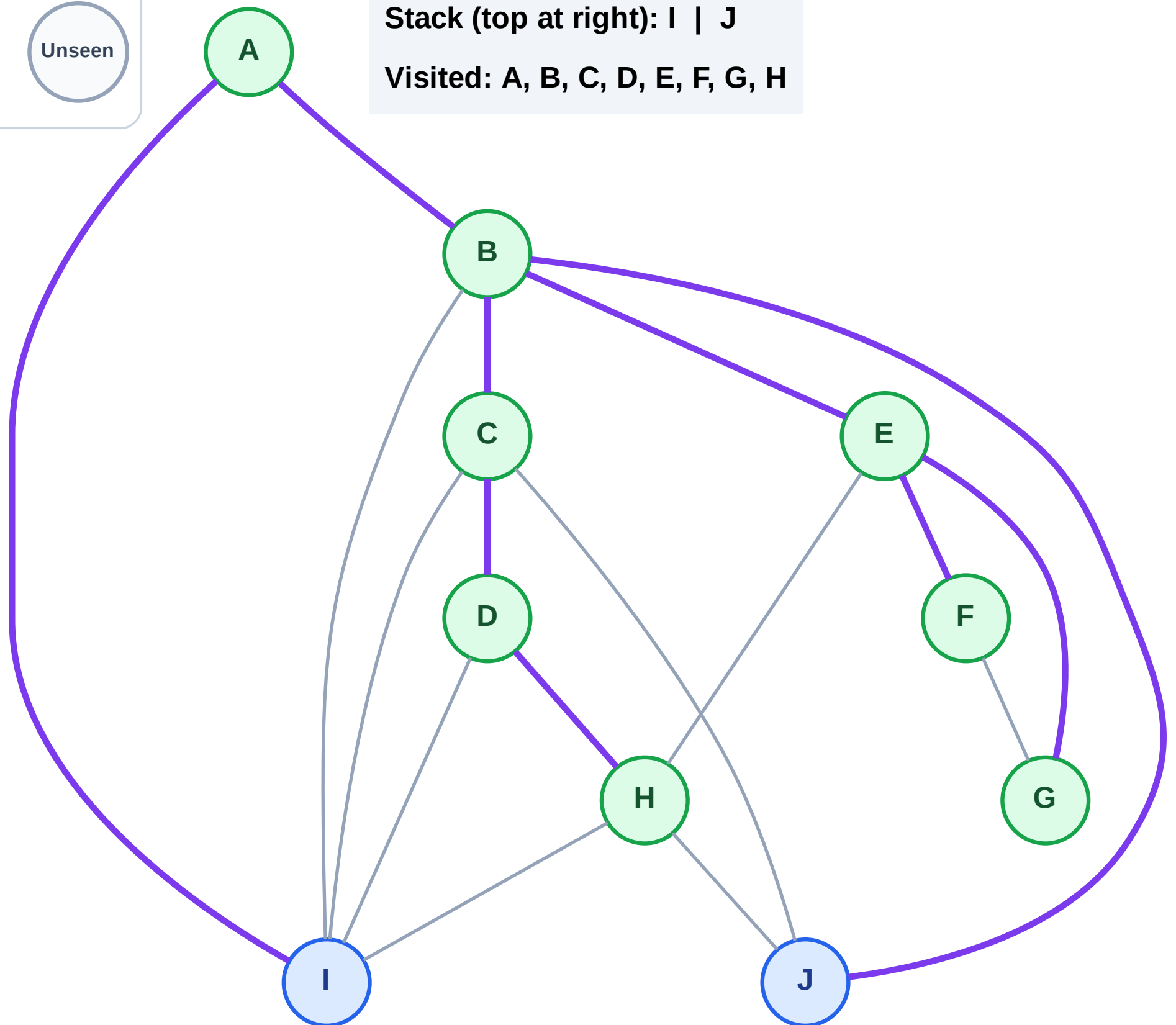
Step 17: No new neighbors from G

Legend



Stack (top at right): I | J

Visited: A, B, C, D, E, F, G, H



DFS Traversal

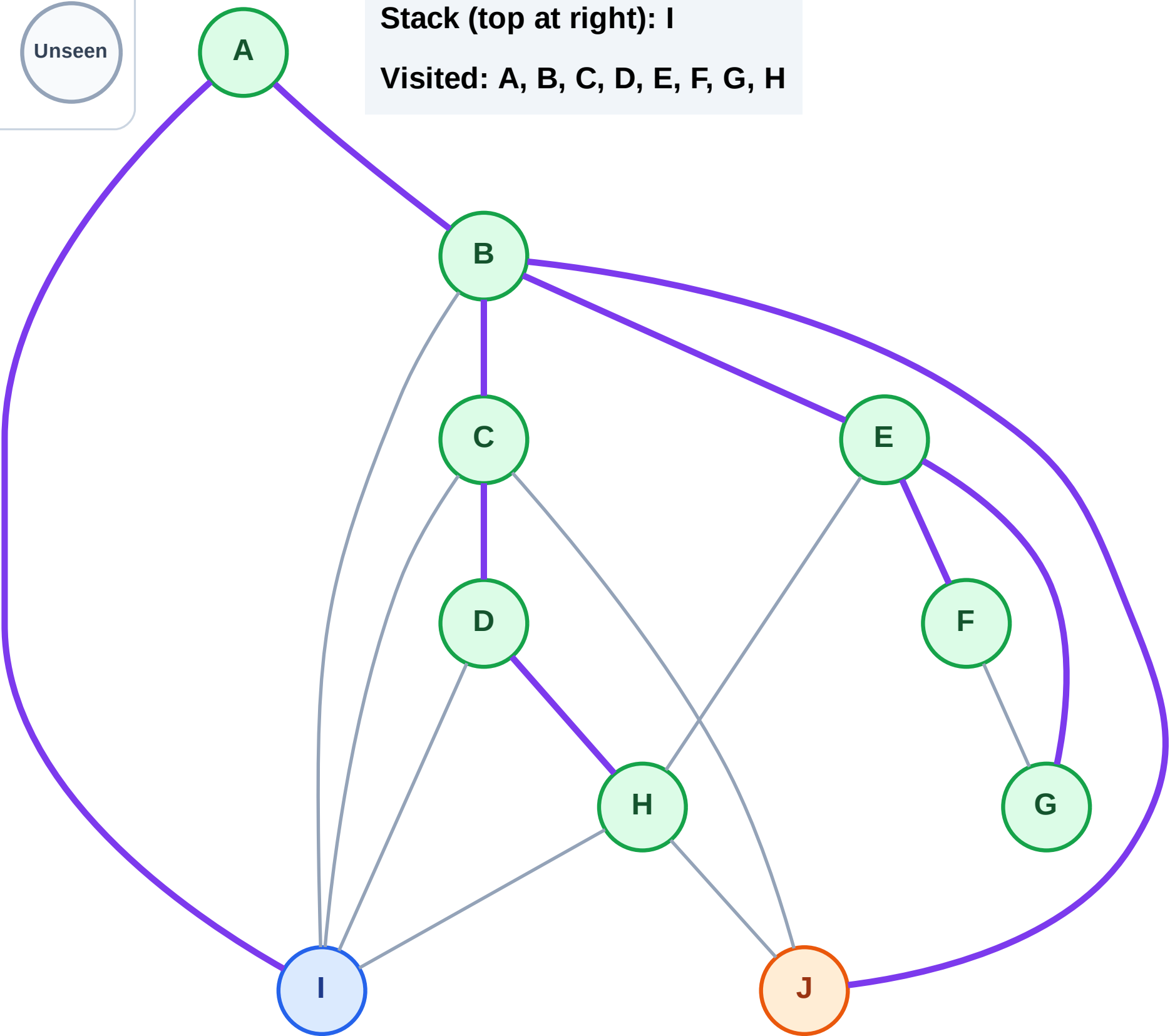
Step 18: Process node J

Legend



Stack (top at right): I

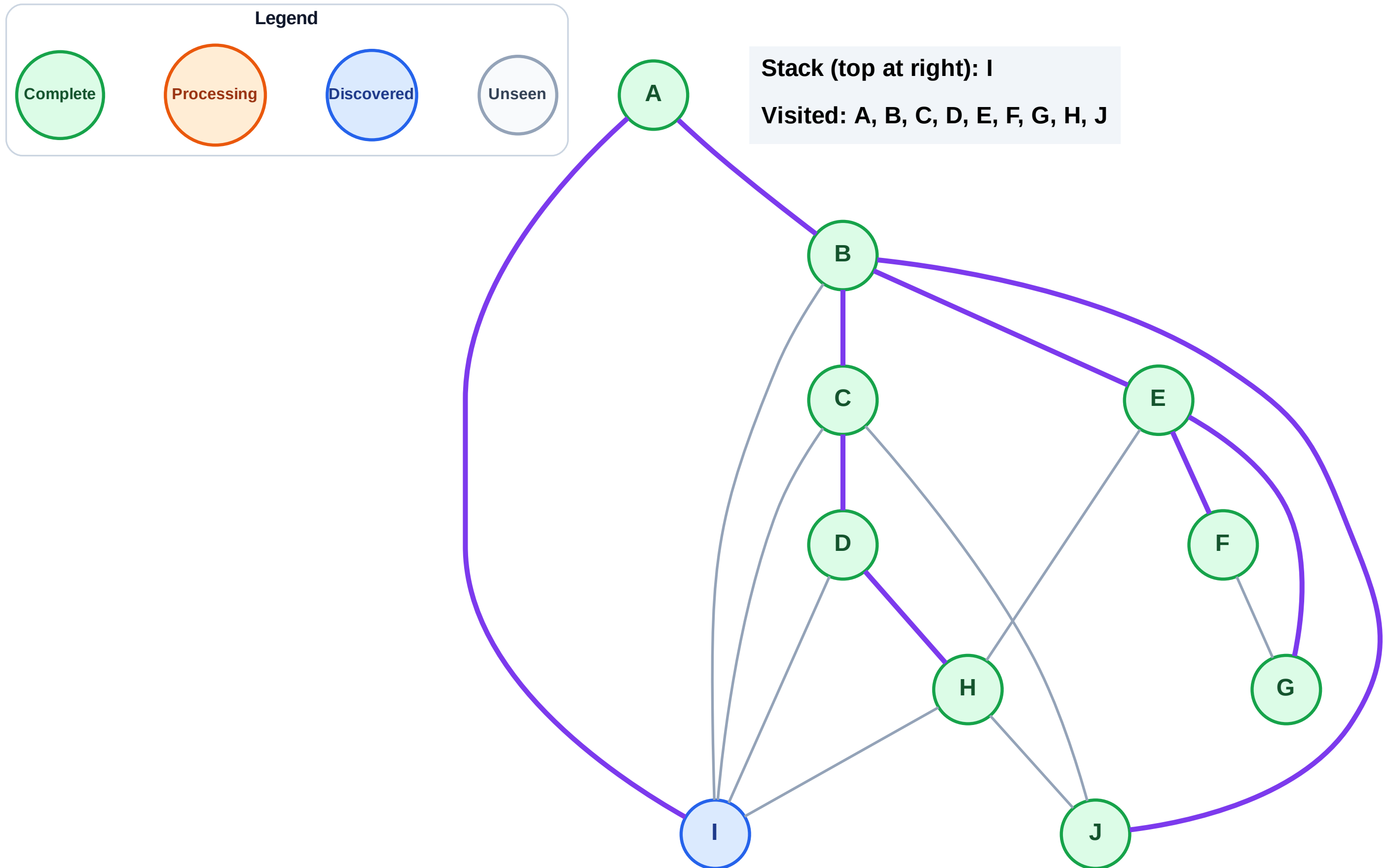
Visited: A, B, C, D, E, F, G, H



Step 19: No new neighbors from J

Stack (top at right): I

Visited: A, B, C, D, E, F, G, H, J



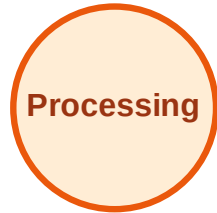
DFS Traversal

Step 20: Process node I

Legend



Complete



Processing



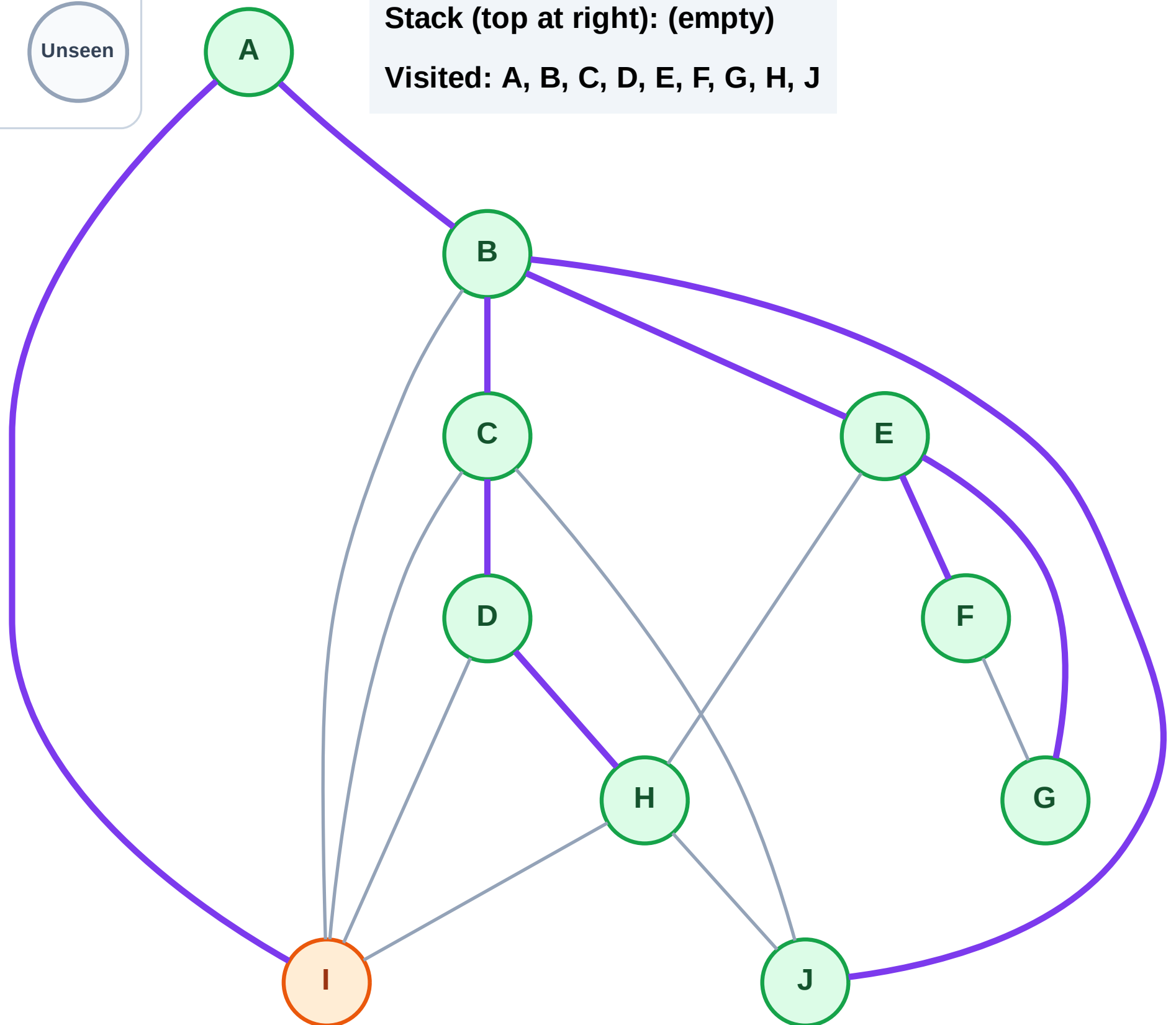
Discovered



Unseen

Stack (top at right): (empty)

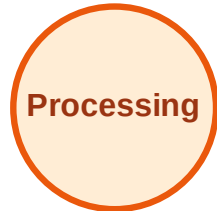
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

